Appendix A of "A two-stage multiobjective optimization algorithm based on estimated Pareto front for handling irregular Pareto fronts"

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ABSTRACT

Multiobjective optimization algorithms based on estimated Pareto front (PF) have achieved promising results. However, the complexity of irregular PFs and insufficient convergence of Pareto nondominated solutions can lead to significant errors in estimating the PF, misguiding the selection of solutions. To address this, we propose a two-stage multiobjective optimization algorithm based on estimated Pareto front (TS-MOEA-EPF). In the first stage, an improved growing neural gas (GNG) guides the population to maintain diversity while converging steadily, thus providing Pareto nondominated solutions close to the true PF. In the second stage, a neural network is used to estimate the PF more flexibly. We then use the projection points of solutions on the estimated PF to more directly and effectively measure the convergence and diversity of solutions, thereby iteratively selecting solutions with good convergence and diversity for the next population one by one. To verify the effectiveness of TS-MOEA-EPF, we conducted comparative experiments with thirteen advanced algorithms on 105 benchmark instances and five practical problems. The results show that TS-MOEA-EPF performs well on most problems, particularly those with irregular PFs or when solutions are far from the true PFs early in the optimization.

A.1. Introduction

This supplementary document includes the following contents:

1. The evaluations of each algorithm's performance on five practical problems (see Section A.2);
2. The analysis of the distribution characteristics of each algorithm on IDTLZ1 and MaF7 (see Section A.3);
3. The comparisons of the actual time costs among the algorithms (see Section A.4);
4. The assessments of the neural network and GNG employed by TS-MOEA-EPF on various problems (see Section A.5);
5. The sensitivity analyses of TS-MOEA-EPF's parameters (see Section A.6);
6. The supplementary experimental data of 13 comparative algorithms and TS-MOEA-EPF (see Section A.7);

A.2. Performance on pratical problems

In this section, MOTSP (multi-objective traveling salesman problem) [1], MOKP (multi-objective knapsack problem) [2], CWDV (crash-worthiness design of vehicles) [3], CSI (car-side impact problem), and WRP (water resource planning problem) [4] are adopted to evaluate the performance of the algorithms, with HV used as the performance metric. The implementation and datasets of MOKP and MOTSP are provided by PlatEMO [5]. When calculating the HV [6], the nadir points used are: $(43, 4.2, 12.5)$ for CSI, (1.7046e+3, 11, 0.3) for CWDV, (8.3061e+4, 1350, 2.8535e+6, 1.5965e+7, 3.4791e+5) for WRP. The ideal points for all problems are set to the origin. For handling constraints, a penalty term is added to each objective, equal to the number of violated constraints multiplied by the maximum value of that objective. This ensuring that feasible solutions dominate those with constraint violations.

To better present the results, the data of TS-MOE-EPF and the 13 comparative algorithms are split into two separate tables. The HV values obtained by each algorithm on these problems are listed in Table A.1 and Table A.2, respectively. TS-MOEA-EPF achieved three best values on MOKP and WRP, while HEA and RVEA-iGNG outperformed TS-MOEA-EPF on MOTSP. Specifically, HEA achieved two best values on MOSTP with 3, 10 objectives, and RVEA-iGNG achieved one best value on the 15-objective MOTSP. CA-MOEA performed best on CWDV, while AdaW and AR-MOEA performed best on 3-objective MOKP and CSI, respectively.

Fig. A.1 shows the distribution of nondominated solutions obtained by representative algorithms for the above problems. From Fig. A.1, it can be observed that TS-MOEA-EPF can maintain good diversity on 3-objective CWDV, CSI and MOKP. HEA and MultiGPO show room for improvement in terms of uniformity on the 5-objective WRP. On MOTSP and MOKP, the distribution of solutions obtained by each algorithm appear to be irregular, which also reflects the complexity of PFs in practical problems. Compared with

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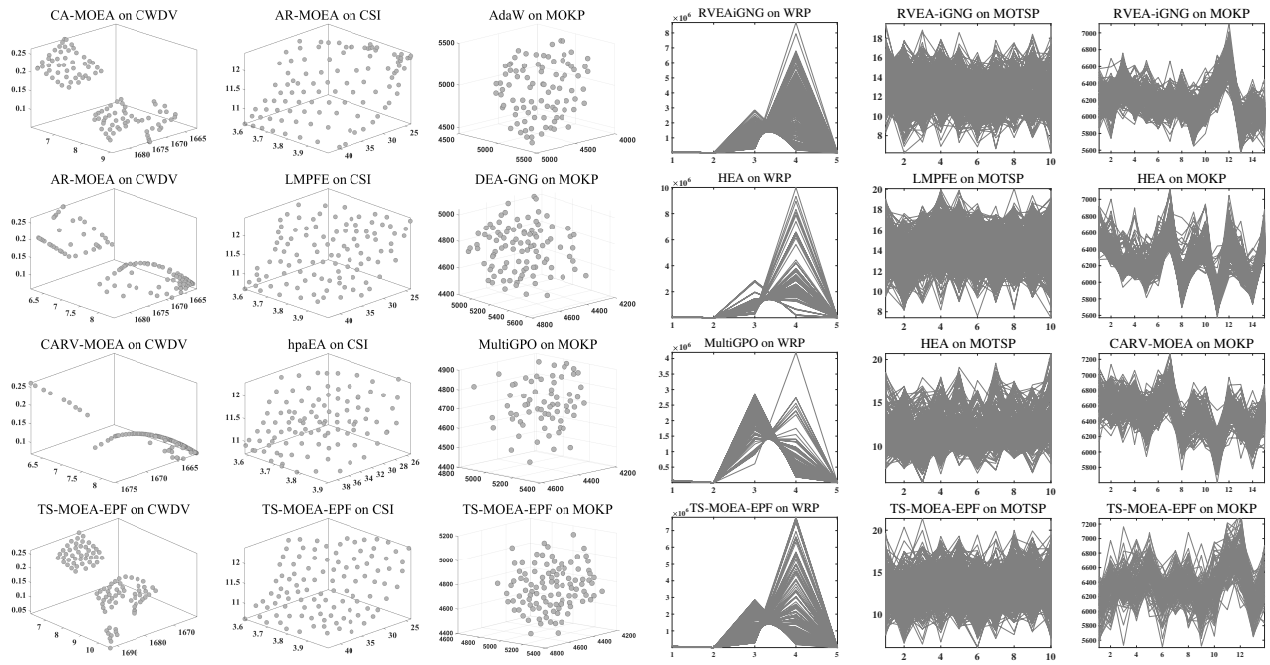

Fig. A.1: The nondominated solutions closest to the HV mean, solved by TS-MOEA-EPF and the comparison algorithms on CWDV, CSI, WRP, MOTSP and MOKP

Table A.1
HV values obtained by TS-MOEA-EPF and 7 of 13 MaOEAs on the practical problems, where "+", "-", and "=" respectively represent the comparison algorithms being superior, inferior, and equivalent to TS-MOEA-EPF (Part I)

Problem	M	CA-MOEA	DEA-GNG	RVEA-iGNG	PeEA	LMPFE	HEA	RVEA-Adak	TS-MOEA-EPF
MOTSP	3	5.6705e-1(1.26e-2)-	5.6261e-1(1.24e-2)-	5.6457e-1(1.08e-2)-	5.6213e-1(9.58e-3)-	5.6951e-1(1.27e-2)=	6.1492e-1(1.37e-2)+	5.3421e-1(1.17e-2)-	5.7810e-1(1.10e-2)
	10	2.1820e-2(9.14e-4)-	3.1046e-2(1.59e-3)-	4.6079e-2(1.13e-3)+	3.3649e-2(1.37e-3)-	2.5726e-2(1.30e-3)-	5.2813e-2(3.07e-3)+	3.0600e-2(1.47e-3)-	4.1138e-2(1.04e-3)
	15	1.4402e-3(9.34e-5)-	2.2458e-3(1.81e-4)-	4.1055e-3(1.60e-4)+	2.3100e-3(1.70e-4)-	2.0509e-3(1.51e-4)-	4.0405e-3(4.53e-4)+	2.5515e-3(1.70e-4)-	3.3889e-3(1.25e-4)
MOKP	3	3.4498e-1(8.81e-3)=	3.3676e-1(9.62e-3)-	3.4376e-1(2.01e-3)-	3.5150e-1(2.25e-3)+	3.4664e-1(2.37e-3)=	3.4178e-1(1.89e-3)-	3.4349e-1(2.18e-3)-	3.4712e-1(2.14e-3)
	10	8.8207e-3(1.79e-4)-	8.3663e-3(2.50e-4)-	9.3384e-3(1.72e-4)-	8.7411e-3(2.24e-4)-	8.9430e-3(1.77e-4)-	9.4078e-3(1.80e-4)-	8.3939e-3(1.78e-4)-	9.8119e-3(1.65e-4)
	15	5.1790e-4(1.78e-5)-	4.9525e-4(3.68e-5)-	6.5205e-4(2.94e-5)=	5.1045e-4(2.78e-5)-	5.3995e-4(2.61e-5)-	5.2080e-4(2.20e-5)-	5.4845e-4(2.79e-5)-	6.6195e-4(2.63e-5)
CWDV	3	4.2597e-2(2.27e-4)=	4.2025e-2(4.40e-4)-	4.1947e-2(1.12e-3)=	4.0216e-2(4.94e-4)-	4.1847e-2(9.20e-4)-	4.1990e-2(4.80e-4)-	3.8241e-2(6.86e-4)-	4.2067e-2(1.07e-3)
CSI	3	1.9920e-2(4.53e-5)-	1.9616e-2(2.28e-4)-	2.0265e-2(6.73e-5)-	1.9748e-2(1.27e-4)-	2.0019e-2(6.22e-5)-	2.0120e-2(4.75e-5)-	1.9405e-2(2.37e-4)-	2.0306e-2(4.03e-5)
WRP	5	2.1127e-1(1.22e-3)-	2.0924e-1(3.54e-3)-	2.1528e-1(1.57e-3)-	2.1101e-1(2.87e-3)-	2.1150e-1(2.93e-3)-	2.1444e-1(8.78e-4)-	2.1325e-1(1.67e-3)-	2.1736e-1(1.95e-3)
+/−/≈		0/7/2	0/9/0	2/5/2	1/8/0	0/7/2	3/6/0	0/9/0	

HEA and CARV-MOEA, RVEA-iGNG and TS-MOEA-EPF cover more objective space on 15-objective MOKP, especially for the 12-th to 15-th objectives.

A.3. Distribution characteristics

From Fig. A.3, the solution distributions obtained by AdaW and TS-MOEA-EPF on IDTLZ1 are comparable to those of other algorithms. To investigate whether their performance on IGD+ and HV is impacted by insufficient convergence or uneven distribution, we evaluate additional metrics: Riesz s-energy E_s [7] for population diversity, GD+ [8] for convergence, and Spacing [9] for uniformity. The calculation of E_s follows Eq. (A.1), where s is set to 2. For all three metrics, smaller values indicate better performance. Furthermore, we use Eq. (A.3) to measure how far the population is from the center and how close it is to the boundary (Boundary Coverage). The results of each indicator are shown in Table A.3.

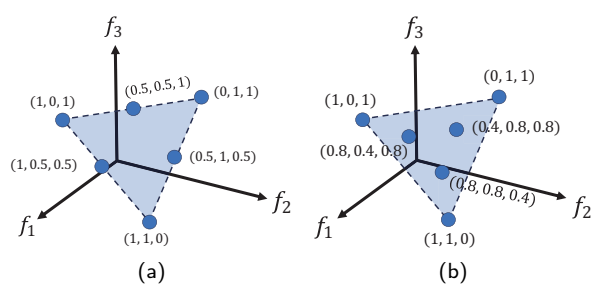

Fig. A.2: (a) These 6 solutions are closer to the boundary, with an IGD+ value of 1.6310e-1 and a HV value of 7.9638e-2, (b) These 6 solutions are closer to the center, with an IGD+ value of 1.3445e-1 and a HV value of 1.1044e-1, both in terms of IGD+ and HV, the latter results are superior to the former

$$E_s(\mathcal{A}) = \sum_{i=1}^{|\mathcal{A}|} \sum_{j=1, j\neq i}^{|\mathcal{A}|} k_s(\vec{a}_i, \vec{a}_j) \tag{A.1}$$

Table A.2
HV values obtained by TS-MOEA-EPF and 6 of 13 MaOEAs on the practical problems, where "+", "-", and "=" respectively represent the comparison algorithms being superior, inferior, and equivalent to TS-MOEA-EPF (Part II)

Problem	M	AdaW	AR-MOEA	hpaEA	MultiGPO	TS-NSGA-II	CARV-MOEA	TS-MOEA-EPF
MOTSP	3	5.6885e-1(1.05e-2)-	5.6575e-1(1.17e-2)-	5.5447e-1(1.08e-2)-	6.0856e-1(1.30e-2)+	5.6188e-1(1.42e-2)-	5.9277e-1(1.68e-2)+	5.7810e-1(1.10e-2)
	10	3.2390e-2(9.75e-4)-	3.3000e-2(1.16e-3)-	2.8833e-2(1.47e-3)-	3.8600e-2(1.91e-3)-	3.1353e-2(1.19e-3)-	3.8485e-2(2.25e-3)-	4.1138e-2(1.04e-3)
	15	2.2680e-3(1.60e-4)-	1.6576e-3(1.55e-3)-	2.6757e-3(1.66e-4)-	2.6028e-3(1.93e-4)-	2.2955e-3(2.68e-4)-	2.9374e-3(4.82e-4)-	3.3889e-3(1.25e-4)
MOKP	3	3.5572e-1(3.74e-3)+	3.2748e-1(1.71e-3)-	3.2525e-1(1.89e-3)-	3.4341e-1(2.71e-3)-	3.4376e-1(4.99e-3)-	3.2603e-1(7.69e-2)-	3.4712e-1(2.14e-3)
	10	9.5742e-3(2.20e-4)-	8.4410e-3(1.62e-4)-	9.0088e-3(1.71e-4)-	9.6027e-3(1.36e-4)-	8.4490e-3(1.86e-4)-	8.8703e-3(1.55e-4)-	9.8119e-3(1.65e-4)
	15	5.8585e-4(2.46e-5)-	5.4705e-4(1.32e-4)-	4.7890e-4(3.30e-5)-	5.1795e-4(1.25e-4)-	5.1620e-4(3.51e-5)-	4.4540e-4(2.50e-5)-	6.6195e-4(2.63e-5)
CWDV	3	4.2425e-2(5.75e-4)=	4.1228e-2(3.84e-4)-	4.0294e-2(6.66e-4)-	4.1135e-2(4.34e-4)-	4.2067e-2(5.99e-4)-	3.9549e-2(8.06e-4)-	4.2067e-2(1.07e-3)
CSI	3	2.0235e-2(6.93e-5)-	2.0312e-2(2.69e-5)=	1.9221e-2(2.43e-4)-	2.0036e-2(9.79e-5)-	1.9980e-2(7.04e-5)-	1.6157e-2(6.45e-4)-	2.0306e-2(4.03e-5)
WRP	5	2.1633e-1(7.74e-4)-	2.1187e-1(1.98e-3)-	2.1190e-1(1.99e-3)-	1.6507e-1(4.15e-2)-	2.1442e-1(8.86e-4)-	2.0332e-1(8.00e-3)-	2.1736e-1(1.95e-3)
+/ − / ≈		1/7/1	0/8/1	0/9/0	1/8/0	0/9/0	1/8/0	

Table A.3
The more metrics value of each algothem on 15-objective IDTLZ1

Problem	GD+	E_s	Spacing	BC
CA-MOEA	2.42e-1(7.6e-1)-	3.54e+7(1.9e+8)-	2.69e+0(5.8e+0)-	1.75e-1(6.0e-3)-
AdaW	1.63e-2(5.5e-2)-	1.17e+5(4.5e+4)-	6.50e-1(2.4e+0)-	1.75e-1(4.15e-3)-
DEA-GNG	4.052e-3(2.6e-4)-	5.20e+7(1.3e+8)-	7.30e-2(1.2e-2)-	1.08e-1(1.6e-2)-
RVEA-iGNG	1.45e-2(4.3e-3)-	3.15e+5(7.4e+4)-	5.25e-2(9.0e-3)=	1.16e-1(1.2e-2)-
PeEA	4.51e-3(9.1e-4)-	1.49e+5(8.2e+4)-	8.07e-2(5.3e-2)-	1.59e-1(1.6e-3)-
LMPFE	4.79e-3(1.4e-3)-	4.57e+5(9.4e+5)-	9.61e-2(3.4e-2)-	1.28e-1(2.5e-2)-
HEA	3.27e-3(6.6e-4)-	6.80e+11(1.2e+12)-	9.47e-2(1.1e-2)-	1.95e-1(1.4e-2)+
RVEA-Ada$\mathcal{K}$	3.96e-2(1.8e-2)-	3.40e+4(6.3e+4)+	2.61e-1(1.8e-1)-	1.87e-1(1.1e-2)=
MultiGPO	3.65e-1(8.3e-1)=	1.33e+5(3.2e+4)-	1.51e+1(3.6e+1)-	1.68e-1(7.7e-3)-
TS-NSGA-II	1.41e-2(3.7e-2)-	1.55e+9(3.4e+9)-	3.67e-1(1.7e+0)-	1.23e-1(1.4e-2)-
AR-MOEA	3.58e-2(8.1e-2)-	2.78e+9(4.5e+9)-	1.46e+0(3.5e+0)-	1.72e-1(6.7e-3)-
hpaEA	3.12e-3(1.2e-4)-	4.08e+5(5.0e+5)-	1.44e-1(7.3e-3)-	1.64e-1(2.5e-3)-
CARV-MOEA	4.20e-3(1.1-4)-	1.81e+5(1.4e+4)-	6.80e-2(8.2e-3)-	1.40e-1(2.7e-3)-
TS-MOEA-EPF	2.71e-3(1.6e-4)	9.17e+4(1.2e+3)	5.37e-2(5.7e-3)	1.83e-1(9.2e-4)

$$k_s(\vec{a}_i, \vec{a}_j) = \begin{cases} \frac{1}{||\vec{a}_i - \vec{a}_j||^s}, & \vec{a}_i \neq \vec{a}_j \\ 0, & \text{otherwise} \end{cases} \tag{A.2}$$

$$BC(P) = \sum_{p \in P} \frac{\angle(\boldsymbol{p}, (1, 1, \ldots, 1))}{|P|} \tag{A.3}$$

From Table A.3, TS-MOEA-EPF demonstrates good convergence, diversity, and uniformity on IDTLZ1. The BC value suggests that TS-MOEA-EPF tends to distribute solutions closer the boundary compared to other algorithms. This indicates that, for inverted Pareto fronts (PFs), boundary solutions contribute less to IGD+ and HV than those near the center, as illustrated in Fig. A.2. Additionally, AdaW and TS-MOEA-EPF exhibit similar values for both E_s and BC, indicating that their distributions on inverted PFs may be very similar. This observation aligns with the similar IGD+ and HV values achieved by AdaW and TS-MOEA-EPF reported in Tables A.9 and shown in Fig. A.3.

From Fig. A.3, the distribution of DEA-GNG on MaF7 is less ideal than those of RVEA-iGNG and TS-MOEA-EPF, yet its HV value is the best. This may be because the solutions obtained by DEA-GNG on MaF7 tend to have larger values on the last objective and smaller values on the first $M-1$ objectives, thus resulting in a solution set that makes a greater contribution to the HV.

A.4. Time Complexity

This section analyzes the actual running time of the compared algorithms and TS-MOEA-EPF under different M and N. For M=3, 10 and 15, the N is set to 105, 275 and 135, respectively. The test problems are: WFG7 and DTLZ6 for 3 objectives, DTLZ7 and IDTLZ2 for 10 objectives, IDTLZ1 and MaF11 for 15 objectives. The maximum function evaluations FE_{max} is set to $M \times 10,000$. All experiments were conducted on PlatEMO [5], running on MATLAB R2022a (Windows 11 Enterprise 64-bit), with hardware specifications: an Intel Core i5-13400 CPU and 32 GB of RAM. Each algorithm was executed 20 times, and the average running time is shown in Fig. A.4.

As shown in Fig. A.4, CA-MOEA, HEA, hpaEA, MultiGPO and TS-NSGA-II are less affected by M and N, achieving relatively lower time costs compared with other algorithms. Meanwhile, AdaW, DEA-GNG, PeEA, AR-MOEA, and RVEA-Ada$\mathcal{K}$ are mainly affected by N: when $N = 275$, their time costs are significantly higher than those at $N = 135$, even though $FE_{\max}$ at $N = 275$ is 2/3 of that at $N = 135$. TS-MOEA-EPF exhibits similar time costs at $M = 10$ ($N = 275$) and $M = 15$ ($N = 135$), indicating that its time cost is affected by M and N. This may be attributed to the process of identifying outliers and extreme points in TS-MOEA-EPF. LMPFE is mainly influenced by M, and as M increases, its time cost gradually increase. This is likely because the time complexity of estimating the PF using polynomial models grows nonlinearly with M. Overall, the time cost of TS-MOEA-EPF is at a moderate level among the compared algorithms.

A.5. Analysis of Neural Network and GNG

across PFs

In this section, we analyze the performance of the neural network and GNG in TS-MOEA-EPF across PFs with varying shapes, M, and N. We use WFG7 (regular), MaF1 and MaF4 (inverted), MaF3 (highly convex), MaF6 (degenerated), MaF7 (disconnected), MaF2 and MaF8 (partial), and MaF9 as representative test problems. It is worth mentioning that the manifold of a degenerated PF has fewer than $M-1$ dimensions. Thus, as M increases, the proportion

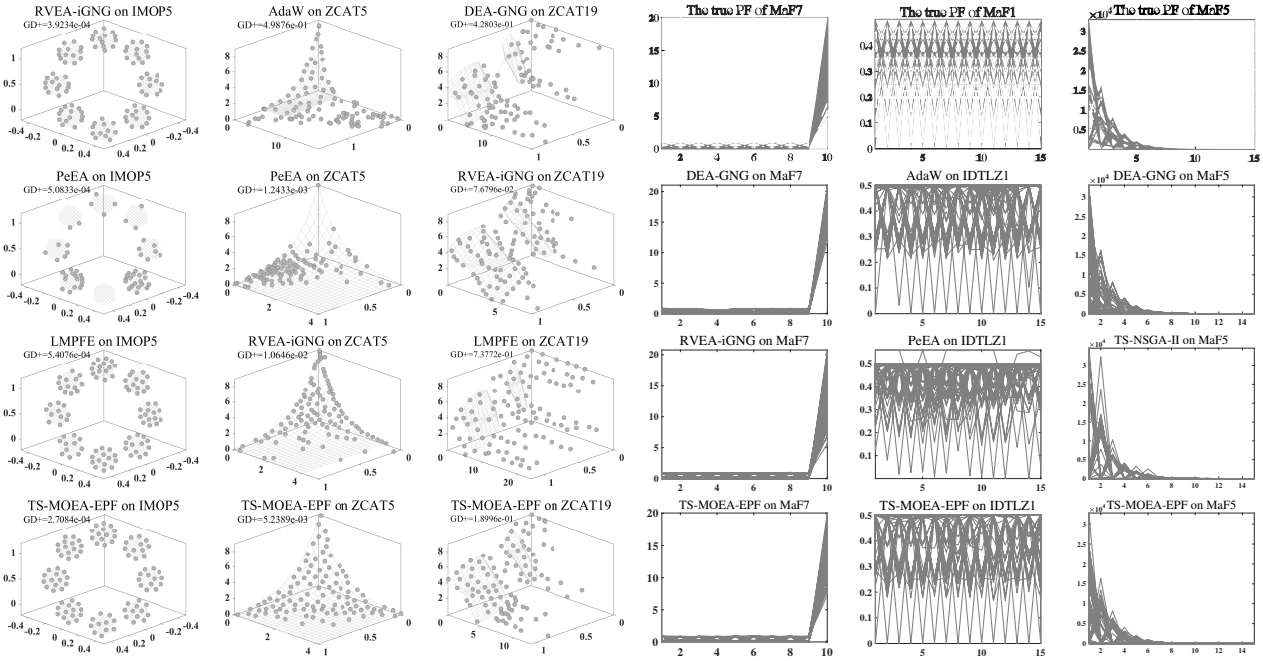

Fig. A.3: The nondominated solutions closest to the IGD+ mean solved by TS-MOEA-EPF and the comparison algorithms

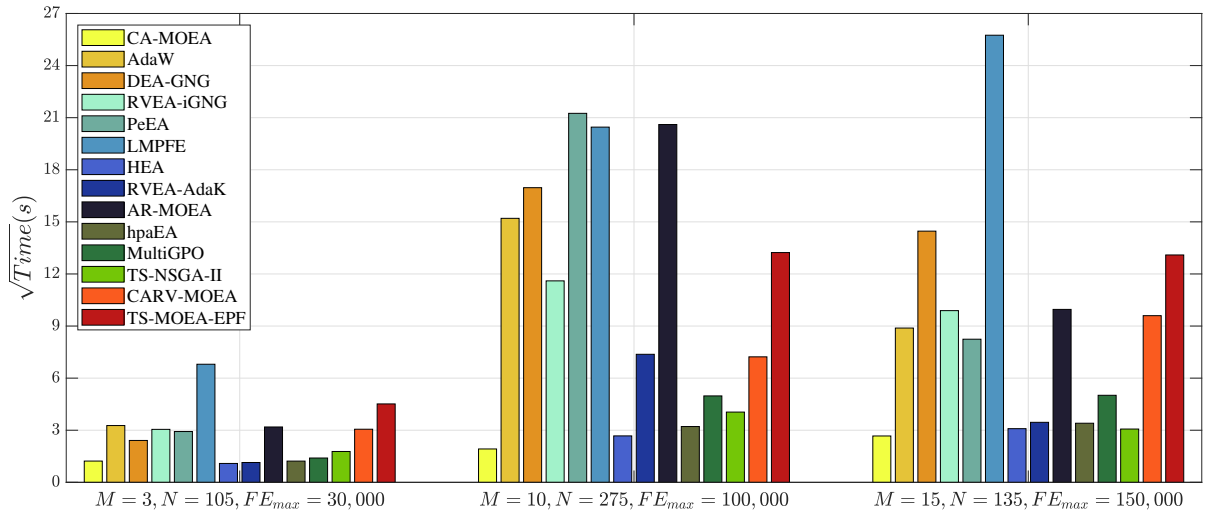

Fig. A.4: The average running time of each algorithm for different M and N

of degenerated PFs in the objective space sharply decreases and becomes increasingly sparse.

A.5.1. Analysis of Neural Network

To measure the fitting ability of the neural network for PFs with varying shapes, N and M, we first normalize the 10,000 uniformly sampled points opt on the true PF using $\boldsymbol{z}^{min}$ and $\boldsymbol{z}^{max}$, obtaining opt'. Here, $\boldsymbol{z}^{min}$ and $\boldsymbol{z}^{max}$ refer to the ideal point and nadir point used to normalize the training set for neural network, which are obtained at 90% of the optimization progress. We then use Eq. (A.4) to compute the corresponding projection points, where $output(\boldsymbol{x})$ denotes the network's output for input $\boldsymbol{x}$. These projection points are restored to the original objective space (before normalization) using Eq. (A.5), followed by re-normalization with the true PF's ideal point $\boldsymbol{z}^{*,min}$ and nadir point $\boldsymbol{z}^{*,max}$. Finally, we use Eq. (A.7) to measure the fitting ability of the neural network. Intuitively, the smaller the MAE_N value, the closer the actual sampling points are to their projected points on the estimated PF.

Meanwhile, to investigate the impact of the number of nondominated solutions on the neural network's ability to approximate the true PF, we executed TS-MOEA-EPF with different population sizes: $0.5N$, N, $2N$, and $3N$.

Specifically, for $M = 3$: $0.5N = 55$, $N = 105$, $2N = 210$, $3N = 300$; for $M = 10$: $0.5N = 110$, $N = 275$, $2N = 440$, $3N = 770$; for $M = 15$: $0.5N = 30$, $N = 135$, $2N = 240$, $3N = 800$. We ensured that FE_{max} remained consistent across different population sizes, adhering to the setting of $M \times 10,000$. The corresponding results are presented in Table A.4. In this table, the value in the last column refers to the mean and variance of the objective sums for opt^*.

$$\boldsymbol{pp}(\boldsymbol{opt}'_i) = \frac{\boldsymbol{opt}'_i}{\sum_{j=1}^{M} opt_{i,j}} \cdot output(\frac{\boldsymbol{opt}'_i}{\sum_{j=1}^{M} opt_{i,j}}) \tag{A.4}$$

$$pp^*_{i,j} = \frac{pp_j(\boldsymbol{opt}'_i) \times (z_j^{max} - z_j^{min}) + z_j^{min} - z_j^{*,min}}{(z_j^{*,max} - z_j^{*,min})} \tag{A.5}$$

$$opt^*_{i,j} = \frac{opt_{i,j} - z_j^{*,min}}{(z_j^{*,max} - z_j^{*,min})} \tag{A.6}$$

$$MAE_N = \frac{1}{|opt^*|} \sum_{i=1}^{|opt^*|} |\frac{\sum_{j=1}^{M} pp^*_{i,j}}{\sum_{j=1}^{M} opt^*_{i,j}} - 1| \tag{A.7}$$

As shown in Table A.4, MAE_N is less than $1e-1$ for all problems, except MaF3 and MaF6 with 10 and 15 objectives, as well as WFG7 and MaF2 with 15 objectives. This suggests the neural network can effectively approximate most PFs. The MAE_N of MaF3 and MaF6 are relatively less ideal compared to other problems. The PF of MaF3 is highly convex: as M increases, the objective sums near its center approach 0, while those near the boundary approach 1. This significant discrepancy in output values easily cause overfitting. For MaF6, the true PF occupies a small portion of the objective space with highly similar input directions but large output differences (large variance of $\sum_{j=1}^{M} opt^*_{\sim,j}$), also leading to overfitting. On the other hand, MAE_N is good for problem with small output differences, such as MaF1 and MaF9, respectively. Therefore, appropriately increasing input diversity while reducing output discrepancy may significantly aid neural network in approximating the true PF.

Regarding the influence of N, except for MaF1, MaF3, and MaF6, MAE_N gradually decreases from $0.5N$ to $2N$ but rises again at $3N$. This may occur because the neural network learns too many noisy samples at $3N$ and suffers from overfitting, or because $3N$ is too large, leading to slow convergence such that the population cannot sufficiently approach the true PF. Based on the Friedman's test [10] at the 0.05 significance level for MAE_N, a population size of $2N$ yields the best approximation to the true PF.

A.5.2. Analysis of GNG

In this section, we analyze and compare the differences in the PF approximation ability of the GNG networks employed by TS-MOEA-EPF and DEA-GNG. We use the average minimum distance between GNG nodes, denoted as $avg(dis)$, to measure the diversity of GNG nodes. Meanwhile, the variance of these minimum distances, denoted as $var(dis)$, is employed to assess the uniformity of GNG nodes. A larger $avg(dis)$ indicates better diversity, while a smaller $var(dis)$ indicates a more uniform distribution.

To measure whether GNG can effectively approximate discontinuous PFs, we introduce θ_G as shown in Eq. (A.10), where NS is the GNG nodes. When a large number of GNG nodes are distributed in discontinuous regions and their objective vectors do not intersect with the true PF, θ_G will be very large. Additionally, we use IGD_G as shown in the Eq. (A.11) to more intuitively determine whether GNG nodes can approximate the true PF. The smaller the IGD_G, the better the node distribution approximate the true PF.

$$NS'_{i,j} = NS_{i,j} \times (z_j^{max} - z_j^{min}) + z_j^{min} \tag{A.8}$$

$$NS^*_{i,j} = \frac{(NS'_{i,j} - z_j^{*,min})}{z_j^{*,max} - z_j^{*,min}} \tag{A.9}$$

$$\theta_G = \frac{\sum_{i=1}^{|opt^*|} \min_{\boldsymbol{r} \in NS^*} \angle(\boldsymbol{opt}^*_i, \boldsymbol{r})}{|opt^*|} \tag{A.10}$$

$$IGD_G = \frac{\sum_{i=1}^{|opt|} \min_{\boldsymbol{r} \in NS'} ||\boldsymbol{opt}_i - \boldsymbol{r}||_2}{|opt|} \tag{A.11}$$

The test problems include regular (WFG7, MaF1), inverted (MaF4), degenerated (MaF6, MaF4), highly convex (MaF3), disconnected (MaF7), partial (MaF2 and MaF8) ones, with 3, 10 and 15 objective. Regarding $avg(dis)$, the above metrics of GNG used by DEA-GNG and TS-MOEA-EPF are shown in Table A.5 (superscript T: TS-MOEA-EPF, superscript D: DEA-GNG). TS-MOEA-EPF shows better node diversity than DEA-GNG, except for MaF3, MaF4, and MaF9. The reason why DEA-GNG exhibits better node diversity on these problems may be attributed to the fact that it also learns and retains outlier nodes that are far from the true PF while still preserving high diversity. However, the IGD_G of DEA-GNG on MaF3, MaF4, and MaF9 is inferior to that of TS-MOEA-EPF, indicating that the nodes of DEA-GNG does not approximate the true PF well.

In terms of $var(dis)$, except for WFG7, MaF2, and MaF6 with 10 and 15 objectives, the node distributions of TS-MOEA-EPF outperform those of DEA-GNG. Fig. A.5 shows the GNG obtained by DEA-GNG and TS-MOEA-EPF. TS-MOEA-EPF outperforms DEA-GNG in uniformity on WFG7, MaF3, MaF4, and MaF7. This may be attributed to the update strategy of the nearest node and its neighboring nodes modified by TS-MOEA-EPF.

In terms of θ_G, TS-MOEA-EPF achieves better θ_G values on discontinuous MaF7 than DEA-GNG, indicating that its nodes are more likely to intersect with the true PF than those of DEA-GNG. This may be because TS-MOEA-EPF adopts a modified definition of the second nearest node and an update method for cumulative errors, which can more accurately record the distribution of nondominated solutions in the objective space compared to DEA-GNG.

Table A.4
MAE_N used to analyze network fitting ability for different population size N

Problem	M	$0.5N$	N	$2N$	$3N$	$\sum_{j=1}^{M} opt^*_{\sim,j}$
WFG7	3	1.3175e-2(3.78e-3)[2]	1.4014e-2(4.03e-3)[2]	1.0364e-2(2.89e-3)[1]	1.0178e-2(4.41e-3)[1]	1.4414e+0(1.69e-1)
	10	6.9178e-2(1.02e-2)[4]	3.8673e-2(5.24e-3)[3]	3.1728e-2(4.88e-3)[1]	3.4519e-2(5.46e-3)[2]	2.0329e+0(4.00e-1)
	15	2.0009e-1(3.00e-2)[3]	1.2743e-1(1.68e-2)[1]	8.5850e-2(1.03e-2)[1]	1.4941e-1(2.55e-2)[2]	2.2547e+0(5.49e-1)
MaF1	3	5.9649e-3(3.34e-3)[1]	6.8344e-3(3.30e-3)[1]	6.8830e-3(4.01e-3)[1]	8.6686e-3(3.21e-3)[1]	2.0000e+0(2.03e-7)
	10	2.5585e-3(1.56e-3)[1]	3.3425e-3(1.71e-3)[1]	5.1726e-3(1.92e-3)[2]	8.3350e-3(1.54e-3)[3]	9.0000e+0(2.82e-6)
	15	1.7049e-3(1.30e-3)[1]	2.1693e-3(1.05e-3)[1]	2.6806e-3(1.15e-3)[2]	6.6963e-3(1.12e-3)[3]	1.4000e+1(5.85e-6)
MaF2	3	7.5693e-3(3.07e-3)[2]	9.2656e-3(3.89e-3)[2]	7.5107e-3(3.23e-3)[1]	1.0773e-2(3.95e-3)[3]	1.4876e+0(7.18e-2)
	10	3.1183e-2(4.54e-3)[3]	1.9662e-2(4.99e-3)[1]	1.5046e-2(3.76e-3)[1]	2.2763e-2(6.62e-3)[2]	3.2312e+0(3.92e-1)
	15	1.1415e-1(4.00e-2)[4]	4.6250e-2(9.24e-3)[3]	3.2802e-2(7.63e-3)[2]	2.3201e-2(7.34e-3)[1]	4.4022e+0(6.54e-1)
MaF3	3	4.3613e-2(8.79e-3)[1]	4.6854e-2(1.21e-2)[1]	7.6649e-2(2.84e-2)[2]	9.5601e-2(3.87e-2)[2]	6.6682e-1(1.25e-1)
	10	2.8284e-1(1.57e-2)[1]	2.7640e-1(2.92e-2)[1]	5.0131e-1(2.38e-1)[2]	2.5227e+1(6.83e+1)[3]	3.1426e-1(1.36e-1)
	15	6.9956e-1(1.72e-1)[3]	3.8188e-1(6.16e-2)[1]	4.6009e-1(1.96e-1)[2]	2.1913e+2(9.50e+2)[4]	2.6425e-1(1.47e-1)
MaF4	3	2.5598e-2(5.35e-3)[1]	2.4488e-2(8.63e-3)[1]	2.6248e-2(9.17e-3)[1]	4.5428e-2(2.39e-2)[2]	1.5586e+0(1.69e-1)
	10	5.9784e-2(1.08e-2)[2]	3.7603e-2(7.02e-3)[1]	3.7330e-2(5.70e-3)[1]	5.2537e-2(1.00e-2)[2]	7.9671e+0(4.00e-1)
	15	1.6008e-1(3.52e-2)[3]	8.3852e-2(1.64e-2)[2]	6.2358e-2(9.79e-3)[1]	6.0298e-2(7.20e-3)[1]	1.2745e+1(5.49e-1)
MaF6	3	6.9205e-3(3.14e-3)[1]	4.9282e-3(3.21e-3)[1]	4.8195e-3(3.00e-3)[1]	8.0151e-3(4.51e-3)[1]	1.8696e+0(3.82e-1)
	10	3.5591e+0(4.27e+0)[1]	4.7279e+0(2.52e+0)[2]	6.3799e+0(3.00e+0)[2]	7.7434e+0(4.97e+0)[2]	6.2321e+0(2.70e+0)
	15	7.4165e-1(7.30e-1)[1]	2.2535e+0(1.33e+0)[2]	2.9077e+0(1.28e+0)[3]	5.6827e+0(3.25e+0)[4]	9.3482e+0(4.37e+0)
MaF7	3	3.6866e-2(5.97e-2)[3]	2.1922e-2(3.45e-2)[2]	1.1027e-2(4.28e-3)[1]	1.6707e-2(4.27e-3)[4]	1.5764e+0(3.19e-1)
	10	5.4999e-2(1.17e-2)[2]	5.0649e-2(7.46e-3)[1]	5.7545e-2(7.94e-3)[2]	7.9071e-2(1.39e-2)[3]	4.4194e+0(1.12e+0)
	15	2.0298e-1(4.25e-2)[4]	1.1020e-1(1.62e-2)[2]	8.2664e-2(1.16e-2)[1]	1.3011e-1(2.45e-2)[3]	7.5000e+0(1.74e+0)
MaF8	3	1.0048e-2(3.17e-3)[1]	9.1259e-3(2.70e-3)[1]	8.5769e-3(3.20e-3)[1]	8.3913e-3(2.77e-3)[1]	1.8235e+0(5.90e-2)
	10	6.3791e-3(9.72e-4)[3]	5.5293e-3(8.87e-4)[1]	5.7128e-3(1.06e-3)[1]	6.1740e-3(1.41e-3)[2]	5.6119e+0(3.69e-1)
	15	2.7508e-2(7.56e-3)[3]	5.8150e-3(4.79e-4)[2]	4.7443e-3(4.51e-4)[1]	4.3433e-3(4.58e-4)[1]	8.4814e+0(5.78e-1)
MaF9	3	8.1106e-3(6.13e-3)[1]	9.2681e-3(4.93e-3)[1]	1.1383e-2(6.37e-3)[1]	1.0784e-2(6.80e-3)[1]	1.0062e+0(2.61e-3)
	10	1.3631e-2(8.44e-3)[1]	1.5161e-2(8.10e-3)[1]	1.3462e-2(8.82e-3)[1]	9.4500e-3(7.09e-3)[1]	5.0000e+0(6.40e-16)
	15	3.6502e-3(3.13e-3)[1]	3.2660e-3(2.76e-3)[1]	2.9656e-3(2.39e-3)[1]	1.4769e-2(1.66e-2)[2]	7.4483e+0(3.96e-3)
Average ranking		2.0000e+0	1.4444e+0	1.3704e+0	2.1111e+0	

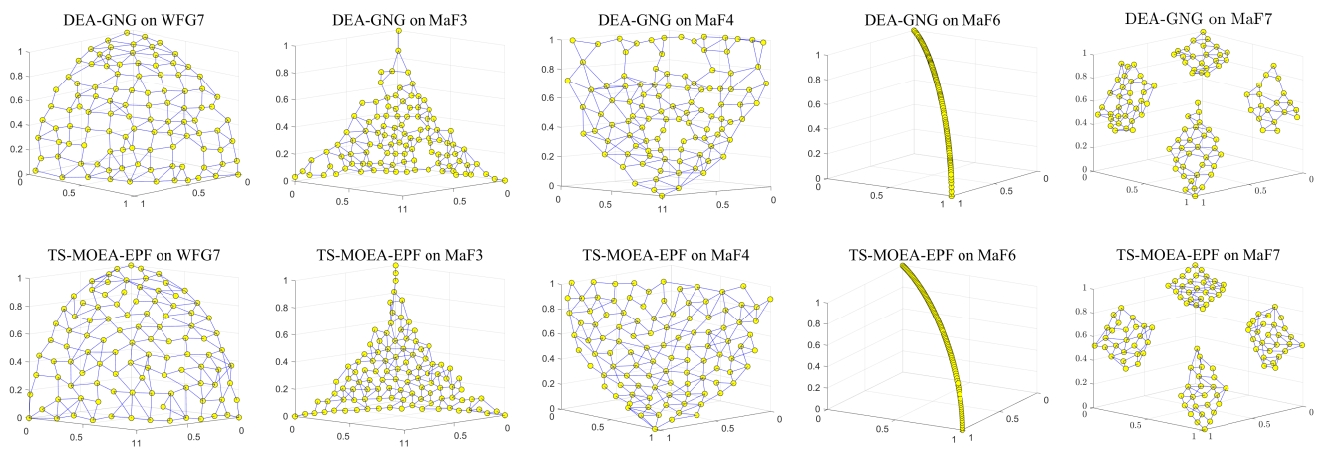

Fig. A.5: The GNG of DEA-GNG and TS-MOEA-EPF on WFG7, MaF3, MaF4, MaF6 and MaF7.

In terms of IGD_G, the nodes of TS-MOEA-EPF are better at approximating the true PF than those of DEA-GNG, except for WFG7 with 3 and 10 objectives, MaF2 with 3 objectives, and MaF6 with 10 and 15 objectives. It should be noted that for MaF6 with 10 and 15 objectives, the GNG node distributions of DEA-GNG and TS-MOEA-EPF are not ideal. This is attributed to their emphasis on maintaining the diversity of the archive. The degenerated PFs are very sparse in the objective space, the points on the true PF dominate only a limited objective space, which further exacerbates the phenomenon of dominance resistance and leads the archive to favor outlier nodes for diversity.

A.6. Parameter analysis

A.6.1. Analysis of θ_1 and θ_2

θ_1 and θ_2 jointly control the sensitivity of the potential outlier detection in Algorithm 3. θ_2 serves as the base threshold, while θ_1 adjusts the detection strictness according to angular differences. Smaller values of θ_1 and θ_2 result in more lenient detection, making solutions more likely to be classified as outliers. Conversely, larger values lead to stricter outlier identification.

To analyze the impact of θ_1 and θ_2 in Algorithm 3, we set c_{div} and c_{cov} to 0.60 and 0.80, respectively, while disabling Algorithm 4. The selected test problems cover various PFs,

Table A.5
The indicator results used to measure the effectiveness of GNG adopted by DEA-GNG (superscript D) and TS-MOEA-EFP (superscript T), where "+", "-", and "=" respectively indicate that the GNG adopted by DEA-GNG is superior to, inferior to, and equivalent to the GNG adopted by TS-MOEA-EPF

Problem	M	$avg^T(dis)$	$avg^D(dis)$	$var^T(dis)$	$var^D(dis)$	θ_G^T	θ_G^D	IGD_G^T	IGD_G^D
WFG7	3	1.251e-1(1.4e-3)	1.043e-1(2.1e-3)-	1.022e-2(2.7e-3)	1.859e-2(1.5e-3)-	4.752e-3(1.6e-4)	4.992e-3(1.7e-4)-	2.614e-2(1.2e-3)	2.555e-2(2.7e-3)=
	10	4.532e-1(8.2e-3)	3.340e-1(2.0e-2)-	7.012e-2(5.2e-3)	5.751e-2(5.8e-3)+	2.160e-1(2.4e-3)	1.973e-1(1.1e-2)+	2.414e+0(3.2e-2)	2.308e+0(1.0e-1)+
	15	5.882e-1(1.4e-2)	3.420e-1(3.3e-2)-	9.867e-2(1.2e-2)	7.203e-2(8.7e-3)+	2.516e-1(4.8e-3)	3.174e-1(2.2e-2)-	4.173e+0(1.2e-1)	6.310e+0(7.9e-1)-
MaF1	3	9.836e-2(1.3e-3)	8.164e-2(1.8e-3)-	4.306e-3(1.7e-3)	1.461e-2(1.7e-3)-	3.584e-3(3.4e-4)	3.289e-3(4.2e-4)+	6.281e-3(7.6e-4)	1.149e-2(8.3e-3)-
	10	3.286e-1(4.8e-3)	1.964e-1(8.7e-3)-	4.100e-2(2.9e-3)	5.193e-2(4.4e-3)-	3.980e-2(7.5e-4)	4.198e-2(2.5e-3)-	1.237e-1(5.4e-3)	2.084e-1(3.5e-2)-
	15	5.472e-1(8.1e-3)	2.657e-1(1.7e-2)-	5.417e-2(1.1e-2)	1.207e-1(1.4e-2)-	4.051e-2(2.7e-3)	5.409e-2(6.7e-3)-	1.934e-1(3.4e-2)	2.835e-1(5.1e-2)-
MaF2	3	1.046e-1(1.7e-3)	8.586e-2(1.9e-3)-	4.877e-3(2.2e-3)	1.544e-2(1.7e-3)-	1.120e-2(9.8e-4)	1.101e-2(1.3e-3)=	8.771e-3(1.7e-3)	8.101e-3(1.5e-3)=
	10	3.142e-1(1.0e-2)	2.082e-1(8.5e-3)-	6.286e-2(4.8e-3)	4.952e-2(4.3e-3)+	1.949e-1(5.9e-3)	2.328e-1(7.6e-3)-	1.239e-1(4.6e-3)	1.501e-1(8.8e-3)-
	15	4.424e-1(1.6e-2)	2.504e-1(2.0e-2)-	1.061e-1(9.3e-3)	5.301e-2(5.2e-3)+	2.425e-1(6.6e-3)	3.208e-1(1.1e-2)-	1.506e-1(6.6e-3)	2.085e-1(1.2e-2)-
MaF3	3	7.346e-2(1.8e-3)	4.820e+3(1.7e+4)=	5.388e-3(2.1e-3)	3.629e+4(1.3e+5)-	4.493e-3(2.0e-4)	5.211e-3(6.7e-4)-	7.873e-3(3.0e-3)	1.077e+5(3.9e+5)-
	10	4.130e-2(4.8e-3)	1.889e+10(6.4e+9)+	1.593e-2(2.8e-3)	7.180e+10(3.0e+10)-	1.951e-1(8.7e-3)	6.112e-2(1.4e-2)+	3.209e-2(2.1e-3)	1.060e+11(4.3e+10)-
	15	4.852e-2(4.5e-3)	8.510e+10(2.1e+10)+	2.243e-2(4.7e-3)	2.091e+11(7.0e+10)-	2.716e-1(1.5e-2)	1.630e-1(2.3e-2)+	3.861e-2(3.2e-3)	2.590e+11(6.3e+10)-
MaF4	3	1.203e-1(7.4e-3)	6.856e-1(1.8e+0)=	1.272e-2(6.5e-3)	2.912e+0(8.2e+0)-	5.347e-3(2.2e-3)	2.007e-2(4.4e-2)-	8.572e-2(8.5e-2)	3.762e+0(8.2e+0)-
	10	3.628e-1(3.9e-2)	8.963e-1(4.4e-1)+	7.464e-2(2.2e-2)	4.859e+0(3.9e+0)-	8.866e-2(1.0e-2)	1.120e-1(2.0e-2)-	5.732e+1(3.4e+1)	6.273e+2(4.6e+2)-
	15	5.923e-1(3.1e-2)	1.393e+0(5.0e-1)+	1.233e-1(3.1e-2)	5.059e+0(2.2e+0)-	8.901e-2(5.5e-3)	1.406e-1(1.9e-2)-	2.637e+3(5.1e+2)	2.982e+4(1.7e+4)-
MaF6	3	1.819e-2(1.2e-4)	1.675e-2(1.8e-4)-	3.430e-3(1.1e-3)	9.510e-3(4.3e-3)-	4.225e-5(2.3e-6)	4.181e-5(2.3e-6)=	7.905e-4(1.9e-4)	2.603e-3(1.4e-3)-
	10	1.916e+1(5.0e+0)	1.049e+1(2.5e+0)-	1.352e+1(4.9e+0)	8.803e+0(4.7e+0)+	8.872e-1(3.4e-2)	9.082e-1(2.3e-2)=	1.516e+2(9.5e+0)	1.452e+2(1.7e+1)=
	15	7.754e+1(1.2e+1)	2.913e+1(9.4e+0)-	5.349e+1(1.9e+1)	1.824e+1(7.7e+0)+	9.375e-1(1.7e-2)	8.935e-1(6.4e-2)=	1.567e+2(9.0e+0)	1.415e+2(2.8e+1)=
MaF7	3	7.062e-2(1.3e-2)	7.165e-2(7.5e-3)=	1.005e-2(2.2e-3)	1.454e-2(1.3e-3)-	5.809e-3(1.4e-3)	6.371e-3(1.3e-3)=	1.820e-2(6.2e-3)	1.857e-2(4.2e-3)=
	10	9.360e-1(2.1e-2)	5.601e-1(9.1e-2)-	3.919e-2(1.5e-2)	2.573e-1(2.9e-2)-	1.342e-1(2.4e-3)	2.109e-1(1.4e-2)-	3.709e-1(1.8e-2)	3.802e+0(1.7e+0)-
	15	1.100e+0(3.3e-2)	4.891e-1(1.5e-1)-	2.067e-1(1.8e-2)	1.807e-1(6.2e-2)+	2.976e-1(2.0e-2)	3.783e-1(4.0e-2)-	8.919e-1(6.4e-2)	6.030e+0(7.8e+0)-
MaF8	3	8.431e-2(3.4e-3)	6.802e-2(2.7e-3)-	6.837e-3(1.7e-3)	2.019e-2(2.8e-3)-	6.561e-3(7.3e-4)	8.024e-3(1.6e-3)-	2.116e-2(9.5e-3)	2.703e-2(1.1e-2)-
	10	1.193e-1(1.1e-3)	9.593e-2(5.7e-3)-	9.958e-3(1.4e-3)	3.484e-2(7.8e-3)-	1.024e-2(3.3e-3)	1.692e-2(4.4e-3)-	4.965e-2(1.1e-2)	7.401e-2(1.8e-2)-
	15	2.146e-1(3.8e-3)	1.759e-1(9.2e-3)-	1.760e-2(4.1e-3)	6.148e-2(1.6e-2)-	2.127e-2(3.4e-3)	2.446e-2(5.1e-3)-	1.175e-1(1.8e-2)	1.265e-1(2.4e-2)=
MaF9	3	9.719e-2(3.6e-3)	8.408e+0(1.3e+1)+	1.095e-2(9.0e-3)	6.091e+1(1.2e+2)-	8.260e-3(3.6e-4)	7.404e-3(5.8e-4)+	1.259e-2(2.3e-3)	8.922e+1(1.6e+2)-
	10	2.996e+0(2.5e+0)	7.416e+1(1.2e+1)+	3.470e+0(3.5e+0)	2.321e+2(9.0e+1)-	4.574e-1(3.9e-2)	5.084e-1(2.1e-3)-	1.296e+2(1.1e+2)	5.142e+3(1.9e+3)-
	15	2.199e-1(6.6e-3)	1.031e+1(9.7e+0)+	2.166e-2(3.1e-2)	6.721e+1(7.0e+1)-	1.767e-2(5.4e-3)	1.533e-1(7.6e-2)-	1.066e-1(1.3e-1)	4.621e+1(4.1e+1)-
+/-/=			7/17/3		7/20/0		5/17/5		1/20/6

Table A.6
The IGD+ for different θ_1 and θ_2 in algorithm 3

Problem	3	No Algorithm 3	θ_1, θ_2 2.0, 1.2	θ_1, θ_2 2.0, 1.5	θ_1, θ_2 3.0,2.0	θ_1, θ_2 3.5, 1.5	θ_1, θ_2 4.0,2.5	θ_1, θ_2 6.0, 3.0
VNT2	3	3.2334e-3(1.55e-4)[2]	4.0276e-3(3.89e-4)[4]	3.2368e-3(2.05e-4)[1]	3.2086e-3(2.02e-4)[1]	3.2028e-3(2.17e-4)[1]	3.3911e-3(2.40e-4)[3]	3.1951e-3(2.60e-4)[1]
MaF5	3	7.7509e-2(1.01e-3)[1]	7.7321e-2(1.01e-3)[1]	7.7209e-2(9.96e-4)[1]	1.0258e-1(1.12e-1)[1]	7.7499e-2(9.90e-4)[1]	1.0214e-1(1.12e-1)[1]	2.5363e-1(3.77e-1)[1]
MaF7	3	5.6797e-2(7.13e-2)[1]	5.7374e-2(7.21e-2)[1]	2.7636e-2(6.33e-4)[1]	3.7604e-2(4.35e-2)[1]	4.7044e-2(5.97e-2)[1]	4.7320e-2(5.99e-2)[1]	3.7982e-2(4.40e-2)[1]
MaF4	10	2.3500e+1(2.10e+1)[5]	1.7443e+1(1.55e+0)[4]	1.8302e+1(2.07e+0)[5]	1.4847e+1(1.40e+0)[2]	1.4561e+1(1.32e+0)[1]	1.5790e+1(1.98e+0)[4]	1.5111e+1(2.24e+0)[3]
MaF7	10	6.9869e-1(3.92e-2)[4]	6.5826e-1(1.95e-2)[1]	6.6335e-1(1.80e-2)[3]	6.6107e-1(1.68e-2)[1]	6.5796e-1(1.60e-2)[1]	6.7024e-1(2.11e-2)[3]	6.6677e-1(2.69e-2)[2]
MaF9	10	5.0394e-1(1.57e-1)[1]	1.6081e+0(2.21e+0)[2]	3.4797e+0(4.09e+0)[5]	5.4213e+0(3.93e+0)[6]	2.4033e+0(3.33e+0)[4]	2.0433e+0(2.53e+0)[3]	3.5180e+0(3.63e+0)[5]
DTLZ6	10	3.0020e-1(1.13e-1)[1]	2.4590e-1(1.40e-1)[1]	3.5266e-1(4.12e-1)[1]	3.0042e-1(2.44e-1)[1]	3.8305e-1(3.61e-1)[1]	2.3356e-1(1.39e-1)[1]	3.6369e-1(4.00e-1)[1]
MaF3	15	6.5159e-1(1.06e+0)[6]	1.9482e-2(3.21e-3)[2]	2.0574e-2(2.57e-3)[2]	1.9076e-2(1.94e-3)[1]	2.6260e-2(6.35e-3)[3]	3.5224e-2(1.28e-2)[4]	4.8189e-2(1.84e-2)[5]
MaF5	15	1.5524e+(3.92e-2)[1]	1.5557e+1(5.13e-2)[1]	1.5514e+1(5.57e-2)[1]	1.5486e+1(3.50e-2)[1]	1.5447e+1(4.32e-2)[1]	1.5895e+1(2.35e-1)[1]	1.5494e+1(4.86e-2)[1]
MaF9	15	2.5808e-1(6.18e-2)[6]	6.9681e-1(2.56e+0)[2]	1.1980e-1(5.46e-4)[2]	1.1991e-1(1.91e-3)[1]	1.2148e-1(3.49e-3)[3]	1.5804e-1(4.10e-2)[4]	2.0682e-1(6.45e-2)[5]
WFG1	15	4.1202e-1(1.22e-1) [6]	3.1952e-1(9.72e-2) [2]	2.8068e-1(5.89e-2)[1]	3.2942e-1(8.12e-2) [4]	3.2024e-1(9.80e-2) [3]	3.3230e-1(7.46e-2) [5]	3.8172e-1(1.15e-1)[7]
WFG2	15	4.4202e-1(1.69e-1)[4]	3.4415e-1(6.45e-2)[1]	3.4027e-1(4.86e-2)[1]	4.1466e-1(1.31e-1)[3]	3.6119e-1(7.66e-2)[2]	4.4897e-1(1.85e-1)[3]	5.0177e-1(2.08e-1)[4]
DTLZ3	15	2.2339e+0(1.57e+0)[7]	3.5923e-1(2.58e-2)[5]	3.2089e-1(1.71e-2)[2]	3.3346e-1(2.77e-2)[3]	3.2692e-1(6.44e-2)[1]	3.5221e-1(2.95e-2)[4]	3.8905e-1(8.42e-2)[6]
Average ranking		3.45615e+0	2.0769e+0	2.0000e+0	1.8462e+0	1.7692e+0	2.8462e+0	3.2308e+0

including the highly convex VNT2; the 10-objective MaF9, DTLZ3, and MaF3, which are prone to encounter outliers; the discontinuous MaF7 and WFG2; the inverted MaF4; the degenerate DTLZ6; and the regular MaF5. These problems involve 3, 10, and 15 objectives, ensuring a comprehensive evaluation of the algorithm's performance across different scenarios. To determine appropriate values for θ_1 and θ_2, they are gradually increased from 2.0 and 1.2 to 6.0 and 3.0, respectively. During this process, their values are further fine-tuned for local optimization. For example, θ_1 is adjusted from 3 to 3.5, and θ_2 from 2 to 1.5.

The IGD+ results in Table A.6 indicate that θ_1 and θ_2 significantly affect performance on problems such as MaF3, MaF4, DTLZ3 and VNT2. For MaF3, MaF4 and DTLZ3,

Table A.7
The IGD+ for different c_{cov} and c_{div} in Algorithm 9

Problem	M	c_{cov}, c_{div} 0.60, 0.90	c_{cov}, c_{div} 0.60, 0.80	c_{cov}, c_{div} 0.60, 0.70	c_{cov}, c_{div} 0.60, 0.60	c_{cov}, c_{div} 0.70, 0.60	c_{cov}, c_{div} 0.80, 0.60	c_{cov}, c_{div} 0.90, 0.60
WFG1	3	5.7115e-2(1.94e-3)[1]	5.8362e-2(2.37e-3)[2]	5.8721e-2(2.18e-3)[3]	6.0177e-2(2.84e-3)[4]	5.9665e-2(1.58e-3)[4]	6.0905e-2(2.64e-3)[4]	6.0635e-2(2.65e-3)[4]
MaF9	3	4.0568e-2(7.88e-4)[1]	4.2242e-2(9.02e-3)[1]	4.0299e-2(5.63e-4)[1]	4.0341e-2(4.37e-4)[1]	4.1474e-2(5.33e-3)[1]	4.0288e-2(3.76e-4)[1]	4.0556e-2(6.74e-4)[1]
IMOP4	3	3.6724e-3(6.09e-5)[1]	3.6840e-3(8.32e-5)[1]	3.6920e-3(1.40e-4)[1]	3.6837e-3(9.70e-5)[1]	3.6910e-3(9.25e-5)[1]	3.6722e-3(9.08e-5)[1]	3.7666e-3(1.59e-4)[1]
IMOP5	3	2.2207e-2(2.03e-4)[1]	2.2106e-2(2.00e-4)[1]	2.2115e-2(1.69e-4)[1]	2.2065e-2(1.73e-4)[1]	2.2080e-2(1.50e-4)[1]	2.2108e-2(1.31e-4)[1]	2.2047e-2(2.18e-4)[1]
IMOP7	3	1.4669e-2(1.97e-4)[5]	1.4599e-2(2.25e-4)[4]	1.4620e-2(1.84e-4)[3]	1.4543e-2(2.62e-4)[2]	1.4510e-2(1.84e-4)[1]	1.4565e-2(1.96e-4)[2]	1.4731e-2(2.55e-4)[6]
WFG7	10	1.1351e+0(2.85e-2)[1]	1.1352e+0(3.03e-2[1])	1.1383e+0(3.47e-2)[1]	1.1222e+0(2.58e-2)[1]	1.1331e+0(2.95e-2)[1]	1.1344e+0(2.82e-2)[1]	1.1356e+0(3.53e-2)[1]
MaF2	10	1.0116e-1(1.43e-3)[2]	1.0066e-1(1.67e-3)[1]	1.0059e-1(1.56e-3)[1]	1.0030e-1(1.66e-3)[1]	9.9718e-2(1.77e-3)[1]	9.9633e-2(1.99e-3)[1]	9.9638e-2(1.53e-3)[1]
MaF7	10	6.7303e-1(3.25e-2)[6]	6.5809e-1(3.21e-2)[6]	6.4553e-1(1.53e-2)[4]	6.4708e-1(1.60e-2)[5]	6.4465e-1(2.76e-2)[3]	6.4284e-1(2.69e-2)[2]	6.2403e-1(1.76e-2)[1]
DTLZ6	10	3.2288e-1(9.18e-2)[1]	3.8216e-1(4.30e-1)[1]	2.2789e-1(1.49e-1)[1]	3.0391e-1(1.10e-1)[1]	3.4757e-1(4.35e-1)[1]	2.9505e-1(1.89e-1)[1]	2.1413e-1(1.35e-1)[1]
MaF8	15	1.0672e-1(2.01e-3)[5]	1.0533e-1(1.56e-3)[4]	1.0545e-1(9.73e-4)[5]	1.0526e-1(1.38e-3)[2]	1.0440e-1(1.22e-3)[1]	1.0534e-1(2.24e-3)[3]	1.0469e-1(1.16e-3)[1]
MaF11	15	3.8364e-1(1.56e-1)[1]	3.4002e-1(6.92e-2)[1]	3.5268e-1(7.55e-2)[1]	3.4000e-1(5.08e-2)[1]	3.2145e-1(5.25e-2)[1]	3.7269e-1(1.13e-1)[1]	3.7029e-1(9.47e-2)[1]
DTLZ3	15	3.1187e-1(1.38e-2)[1]	3.3448e-1(9.25e-2)[1]	3.0535e-1(2.77e-2)[1]	3.1538e-1(2.39e-2)[1]	3.3366e-1(1.20e-1)[1]	3.0057e-1(1.82e-2)[1]	3.1176e-1(2.79e-2)[1]
Average ranking		2.0000e+0	1.8462e+0	1.7692e+0	1.5385e+0	1.3077e+0	1.46154e+0	1.5385e+0

extreme points are often selected from outliers with poor convergence, Algorithm 3 can detect such solutions and avoid misclassifying them as extreme points. In contrast, for VNT2 with highly convex PF, small θ_1 and small θ_2 value (such as $\theta_1 = 2$ and $\theta_2 = 1.5$) may incorrectly classify true Pareto nondominated solutions as outliers. Therefore, θ_1 and θ_2 should not be too small. When $\theta_1 \geq 3.5$ and $\theta_2 \geq 1.5$, Algorithm 3 exhibits no adverse effects on VNT2. Meanwhile, the above parameter settings have little impact on problems with regular PFs (e.g., MaF5).

Based on the Friedman's test at the 0.05 level and the average ranking shown in Table A.6, we set θ_1 to 3.5 and θ_2 to 1.5, respectively.

A.6.2. Analysis of c_{cov} and c_{div}.

This section primarily evaluates the performance of c_{cov} and c_{div} in balancing diversity and convergence across various PFs. c_{cov} and c_{div} respectively determine the weights of convergence and diversity in Algorithm 9. Larger c_{cov} reduces the scaled differences in convergence among solutions and lowers its weight in fitness evaluation; similarly, larger c_{div} reduces the scaled differences in diversity and decreases its corresponding weight. Thus, the algorithm prioritizes convergence when c_{cov} is small and c_{div} is large, and favors diversity otherwise. In the parameter settings, we start with $c_{cov} = 0.6$, $c_{div} = 0.9$ and gradually adjust to $c_{cov} = 0.9, c_{div} = 0.6$. The former parameter setting favors convergence, while the latter setting emphasizes diversity.

For test problems, we selected test problems that are less prone to encounter outliers, including regular WFG7, MaF2 and MaF8 with partial PFs, discontinuous MaF7 and MaF11, degenerated IMOP4 and DTLZ6, with M ranging from 3 to 15. For the sensitivity analysis of c_{cov} and c_{div}, we set θ_1 to 3.5 and θ_2 to 1.5, respectively, disabling Algorithm 4. The IGD+ results in Table A.7 show that TS-MOEA-EPF is not highly sensitive to c_{cov} and c_{div}. Diversity should be emphasized more than convergence for MaF7 and MaF8. Because smaller c_{div} (c_{cov}) tends to amplify differences in solution diversity (convergence), and on MaF7 and MaF8 the performance with $c_{cov} = 0.6$ and $c_{div} = 0.9$ is inferior to that with $c_{cov} = 0.9$ and $c_{div} = 0.6$.

Based on the Friedman's test at the 0.05 level and the average rankings, we set c_{cov} and c_{div} to 0.70 and 0.60, respectively.

A.6.3. Analysis of p, ϵ_1 and ϵ_2.

p, ϵ_1 and ϵ_2 control how strictly potential outliers are detected by Algorithm 4. Specifically, smaller p and ϵ_2, coupled with a larger ϵ_1, make Algorithm 4 more lenient in potential outlier detection, making it more likely to classify solutions as outliers. In the parameter experiments, we test three values of p: 1, 1.5 and 2. For each p, we evaluate two combinations of ϵ_1 and ϵ_2: $\epsilon_1 = 5e\text{-}3$, $\epsilon_2 = 7.5e\text{-}2$ and $\epsilon_1 = 1e\text{-}2$, $\epsilon_2 = 1.5e\text{-}1$.

For test problems, we select MaF3, the 10-objective MaF9, and ZCAT problem sets, which are prone to encounter outliers. Additionally, to analyze performance on other problems where outliers are less likely to occur, we include the highly convex VNT2, discontinuous MaF7 and MaF11, and regular WFG7 in the test problems. For the sensitivity analysis of p, ϵ_1, and ϵ_2, we set $\theta_1 = 3.5$ and $\theta_2 = 1.5$, $c_{cov} = 0.70$, and $c_{div} = 0.60$. The IGD+ results are shown in Table A.8. Algorithm 4 has a significant impact on 3-objective ZCAT, performing best when $p = 1$, $\epsilon_1 = 5e\text{-}3$, and $\epsilon_2 = 7.5e\text{-}2$. This setting imposes no negative effects on problems including 15-objective highly convex MaF3, 15-objective discontinuous MaF7, and 15-objective regular WFG7, while achieving competitive performance across other problems. Hence, we set $p = 1$, $\epsilon_1 = 5e\text{-}3$, and $\epsilon_2 = 7.5e{-}2$.

A.7. The supplementary data

This section presents the HV values of 13 comparative algorithms—namely CA-MOEA [12], , DEA-GNG [13], RVEA-iGNG [14], PeEA [15], LMPFE [16], HEA [17], RVEA-AdaK [18], AdaW [19], MultiGPO [20], TS-NSGA-II [21], AR-MOEA [22], hpaEA [23], and CARV-MOEA [24]—and the designed TS-MOEA-EPF on selected test problems from the WFG [25], MaF [26], DTLZ [27], IDTLZ [28], VNT [29], IMOP [30], and ZCAT [30] suites. The corresponding results are shown in Tables A.9 to A.14.

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Table A.8
The IGD+ for different P, ϵ_1 and ϵ_2 in algorithm 4

Problem	M	No Algorithm 4	p, ϵ_1, ϵ_2 1,5e-3,7.5e-2	p, ϵ_1, ϵ_2 1,1e-2,1.5e-1	p, ϵ_1, ϵ_2 1.5,5e-3,7.5e-2	p, ϵ_1, ϵ_2 1.5,1e-2,1.5e-1	p, ϵ_1, ϵ_2 2,5e-3,7.5e-2	p, ϵ_1, ϵ_2 2,1e-2,1.5e-1
VNT2	3	3.6917e-3(2.36e-4)[1]	3.6192e-3(2.32e-4)[1]	3.6093e-3(2.55e-4)[1]	3.6779e-3(3.90e-4)[1]	3.6832e-3(2.47e-4)[1]	3.7136e-3(3.43e-4)[1]	3.5866e-3(2.94e-4)[1]
MaF9	10	2.2762e+0(3.73e+)[5]	6.1993e-1(1.45e-1)[2]	7.1564e-1(1.41e-1)[2]	6.3396e-1(2.03e-1)[1]	7.6519e-1(2.83e-1)[3]	6.1600e-1(2.10e-1)[1]	7.1677e-1(1.73e-1)[4]
MaF3	15	2.3440e-2(5.56e-3)[1]	2.3132e-2(4.18e-3)[1]	2.2253e-2(5.65e-3)[1]	2.3378e-2(9.84e-3)[1]	2.5517e-2(1.30e-2)[1]	2.1434e-2(4.91e-3)[1]	2.2061e-2(3.48e-3)[1]
MaF7	15	1.4299e+0(1.81e-1)[1]	1.4991e+0(2.18e-1)[1]	1.4670e+0(1.90e-1)[1]	1.5658e+0(2.85e-1)[1]	1.4295e+0(1.67e-1)[1]	1.5233e+0(3.19e-1)[1]	1.4925e+0(1.70e-1)[1]
MaF11	15	3.2980e-1(6.72e-2)[1]	3.2644e-1(5.00e-2)[1]	3.1294e-1(5.87e-2)[1]	3.6466e-1(8.01e-2)[1]	3.3636e-1(6.97e-2)[1]	3.4132e-1(7.12e-2)[1]	3.1759e-1(5.00e-2)[1]
WFG7	15	1.5322e+0(4.56e-2)[2]	1.5083e+0(3.91e-2)[1]	1.5586e+0(9.58e-2)[2]	1.5512e+0(2.49e-2)[3]	1.5422e+0(3.08e-2)[2]	1.5414e+0(4.02e-2)[2]	1.5693e+0(1.48e-1)[2]
ZCAT6	2	4.0349e-2(4.01e-2)[1]	3.6857e-2(4.21e-2)[1]	2.7947e-2(3.35e-2)[1]	4.3710e-2(3.64e-2)[1]	5.3147e-2(4.21e-2)[1]	4.1798e-2(3.69e-2)[1]	2.7986e-2(3.65e-2)[1]
ZCAT10	2	3.0039e-2(3.07e-2)[1]	4.5413e-2(3.72e-2)[1]	4.2035e-2(3.84e-2)[1]	4.0816e-2(3.90e-2)[1]	3.6536e-2(3.50e-2)[1]	2.9611e-2(3.40e-2)[1]	4.1777e-2(5.27e-2)[1]
ZCAT11	2	2.1595e-2(4.82e-2)[1]	5.0385e-2(8.18e-2)[1]	4.4067e-2(7.96e-2)[1]	5.8630e-2(8.62e-2)[1]	3.9523e-2(6.92e-2)[1]	2.9836e-2(5.20e-2)[1]	2.0446e-2(4.13e-2)[1]
ZCAT16	2	2.5874e-2(3.89e-2)[1]	3.4593e-2(5.66e-2)[1]	3.7974e-2(6.66e-2)[1]	2.7041e-2(5.41e-2)[1]	4.1159e-2(6.65e-2)[1]	3.5009e-2(6.41e-2)[1]	4.0029e-2(7.66e-2)[1]
ZCAT18	2	1.5708e-1(1.39e-1)[1]	1.5126e-1(1.62e-1)[1]	1.2542e-1(1.59e-1)[1]	1.3391e-1(1.55e-1)[1]	7.2914e-2(1.22e-1)[1]	1.1359e-1(1.44e-1)[1]	1.2536e-1(1.43e-1)[1]
ZCAT19	2	1.8332e-2(1.61e-2)[1]	1.7488e-2(1.73e-2)[1]	3.1843e-2(4.16e-2)[1]	1.4117e-2(1.48e-2)[1]	1.6363e-2(1.32e-2)[1]	9.8604e-3(9.86e-3)[1]	1.4055e-2(1.34e-2)[1]
ZCAT1	3	9.4516e-2(2.32e-2)[1]	8.8564e-2(2.27e-2)[1]	8.8671e-2(2.06e-2)[1]	8.6611e-2(2.56e-2)[1]	8.7770e-2(1.56e-2)[1]	9.3327e-2(1.67e-2)[1]	8.7855e-2(1.73e-2)[1]
ZCAT5	3	5.9548e-2(1.08e-2)[2]	7.0505e-2(1.34e-2)[4]	6.3123e-2(1.18e-2)[2]	6.1950e-2(9.05e-3)[2]	5.9561e-2(8.50e-3)[2]	6.7911e-2(1.32e-2)[3]	6.1040e-2(1.17e-2)[1]
ZCAT7	3	1.7427e-1(3.33e-2)[5]	1.3276e-1(8.92e-2)[1]	1.6013e-1(7.05e-2)[5]	1.2323e-1(8.49e-2)[2]	1.3910e-1(7.23e-2)[3]	1.3343e-1(6.19e-2)[3]	1.5103e-1(7.67e-2)[4]
ZCAT8	3	2.5886e-1(6.77e-2)[4]	1.0046e-1(3.42e-2)[1]	1.2890e-1(6.52e-2)[3]	1.3836e-1(4.85e-2)[3]	1.4117e-1(5.18e-2)[3]	1.3020e-1(4.46e-2)[3]	1.1582e-1(4.26e-2)[2]
ZCAT9	3	1.3059e-1(2.98e-2)[3]	9.0510e-2(1.43e-2)[1]	9.5367e-2(2.04e-2)[3]	9.9579e-2(1.89e-2)[3]	9.5751e-2(2.08e-2)[2]	9.7900e-2(2.37e-2)[3]	1.0353e-1(2.26e-2)[3]
ZCAT10	3	1.6456e-1(2.93e-2)[7]	7.9119e-2(1.65e-2)[1]	8.8646e-2(1.79e-2)[3]	8.7659e-2(1.59e-2)[2]	9.6469e-2(2.05e-2)[5]	1.0071e-1(1.79e-2)[6]	9.4373e-2(1.50e-2)[4]
ZCAT12	3	2.1914e-1(1.40e-1)[5]	1.0641e-1(2.54e-2)[1]	1.1488e-1(2.79e-2)[2]	1.2876e-1(3.34e-2)[4]	1.2354e-1(2.59e-2)[4]	1.2242e-1(3.50e-2)[3]	1.2079e-1(2.57e-2)[3]
ZCAT14	3	2.9455e-1(6.69e-2)[5]	1.0737e-1(3.96e-2)[1]	1.4658e-1(3.48e-2)[4]	1.2353e-1(3.65e-2)[2]	1.4640e-1(3.24e-2)[5]	1.3953e-1(3.55e-2)[3]	1.7106e-1(3.44e-2)[5]
ZCAT15	3	4.6180e-1(1.61e-1)[5]	1.8494e-1(1.16e-1)[1]	2.1164e-1(6.29e-2)[4]	1.7606e-1(5.97e-2)[2]	2.2532e-1(9.43e-2)[4]	2.2265e-1(1.06e-1)[3]	2.3437e-1(1.11e-1)[5]
ZCAT18	3	1.6364e-1(5.90e-2)[4]	9.3584e-2(3.06e-2)[1]	1.2553e-1(6.98e-2)[3]	1.1848e-1(6.44e-2)[2]	1.2427e-1(5.89e-2)[4]	1.2114e-1(4.97e-2)[4]	1.0974e-1(4.23e-2)[3]
ZCAT20	3	1.4396e-1(7.97e-2)[4]	1.0789e-1(9.47e-2)[1]	1.2554e-1(1.05e-1)[2]	1.8252e-1(1.40e-1)[3]	1.5474e-1(1.26e-1)[3]	1.1963e-1(1.02e-1)[1]	1.9318e-1(1.70e-1)[4]
Average ranking		2.6957e+0	1.1739e+0	2.0000e+0	1.8261e+0	2.2174e+0	2.0000e+0	2.2174e+0

Table A.9
HV values obtained by TS-MOEA-EPF and 7 of 13 MaOEAs, where "+", "-", and "=" respectively represent the comparison algorithms being superior, inferior, and equivalent to TS-MOEA-EPF (Part I)

Property	Problem	M	CA-MOEA	DEA-GNG	RVEA-iGNG	PeEA	LMPFE	HEA	RVEA-Ada$\mathcal{K}$	TS-MOEA-EPF
Inverted	IDTLZ1	3	2.1680e-1(2.90e-3)-	2.0036e-1(1.51e-2)-	2.2037e-1(2.39e-3)-	2.1723e-1(1.25e-3)-	1.9775e-1(2.97e-2)-	1.9782e-1(3.62e-2)-	2.1397e-1(3.89e-3)-	2.2127e-1(3.18e-3)
		10	2.8398e-7(5.12e-7)=	5.9519e-7(2.11e-7)+	3.3016e-7(1.98e-7)-	5.9958e-7(5.29e-7)=	4.4401e-7(2.92e-7)=	2.3277e-7(1.20e-7)=	3.8637e-8(1.20e-7)=	3.5512e-7(5.43e-7)
		15	0.0000e+0(0.00e+0)=	4.2329e-12(3.62e-12)+	6.5483e-14(1.76e-13)+	0.0000e+0(0.00e+0)=	7.4654e-12(3.54e-11)+	1.3526e-12(1.04e-12)+	6.8693e-14(2.49e-13)+	0.0000e+0(0.00e+0)
	IDTLZ2	3	5.2775e-1(1.62e-3)-	5.3561e-1(3.46e-3)-	5.4035e-1(7.08e-4)-	5.2636e-1(3.85e-3)-	5.3239e-1(1.45e-3)-	5.2165e-1(8.15e-4)-	5.3242e-1(8.70e-4)-	5.4184e-1(4.47e-4)
		10	1.0258e-6(1.05e-6)-	4.0417e-4(8.69e-5)+	1.8697e-4(1.69e-5)+	5.4044e-4(2.02e-5)+	1.2561e-4(5.05e-6)+	3.1470e-4(2.43e-5)+	2.9383e-5(1.08e-5)-	9.1574e-5(1.97e-5)
		15	0.0000e+0(0.00e+0)=	8.6925e-8(4.02e-8)+	2.5331e-8(8.77e-9)+	3.1046e-7(3.00e-8)+	1.6534e-7(1.29e-7)+	1.4697e-7(1.01e-8)+	4.1040e-9(8.96e-9)+	0.0000e+0(0.00e+0)
	MaF1	3	2.1390e-1(1.15e-3)-	2.2092e-1(6.53e-4)-	2.2214e-1(3.25e-4)-	2.1926e-1(1.51e-3)-	2.1769e-1(9.40e-4)-	2.0660e-1(3.80e-4)-	2.1258e-1(1.46e-3)-	2.2396e-1(1.88e-4)
		10	1.3067e-7(3.39e-7)=	6.0185e-7(2.67e-7)+	3.2782e-7(1.78e-7)+	5.8540e-7(4.57e-7)+	1.5947e-7(3.03e-7)=	6.7220e-7(8.51e-7)+	1.3604e-8(2.35e-8)-	2.3331e-7(4.30e-7)
		15	0.0000e+0(0.00e+0)=	1.3042e-12(3.38e-12)+	7.3824e-14(1.49e-13)+	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	1.5121e-12(4.80e-13)+	2.5998e-14(2.31e-14)+	0.0000e+0(0.00e+0)
	MaF4	3	5.2698e-1(7.73e-3)-	5.1441e-1(3.02e-2)-	5.3560e-1(2.98e-3)+	4.9317e-1(9.38e-2)-	4.3956e-1(7.99e-2)-	5.1733e-1(2.99e-3)-	5.0180e-1(9.98e-2)-	5.3186e-1(5.82e-3)
		10	1.5335e-6(2.23e-6)-	2.2557e-4(8.75e-5)+	2.8523e-4(4.53e-5)+	2.5661e-4(1.38e-5)+	7.6117e-5(1.24e-4)-	2.8332e-4(3.86e-5)+	2.3665e-5(2.04e-5)-	8.6946e-5(2.02e-5)
		15	1.2905e-15(7.07e-15)=	8.9831e-8(2.38e-8)+	9.6719e-8(6.48e-8)+	1.4176e-7(3.26e-8)+	1.5917e-7(1.41e-8)+	9.5289e-9(3.68e-8)+	1.8464e-10(3.07e-10)+	0.0000e+0(0.00e+0)
Partial	MaF2	3	2.4024e-1(8.58e-4)-	2.4234e-1(1.24e-3)-	2.4603e-1(3.53e-4)=	2.4567e-1(4.77e-4)-	2.4423e-1(9.87e-4)-	2.4510e-1(1.42e-4)-	2.3699e-1(1.95e-3)-	2.4618e-1(3.22e-4)
		10	1.9075e-1(4.48e-3)-	1.8747e-1(6.00e-3)-	2.2317e-1(2.16e-3)+	2.0985e-1(4.16e-3)+	1.7179e-1(7.15e-3)-	2.0530e-1(1.28e-3)+	2.0739e-1(2.29e-3)+	2.0198e-1(6.25e-3)
		15	1.0665e-1(8.11e-3)-	1.7201e-1(8.17e-3)+	1.9653e-1(4.06e-3)+	2.1625e-1(2.60e-3)+	1.0534e-1(2.03e-2)-	1.6975e-1(3.80e-3)+	1.3908e-1(7.57e-3)-	1.5555e-1(9.68e-3)
	MaF8	3	2.7760e-1(8.02e-4)-	2.7015e-1(4.53e-3)-	2.8067e-1(5.15e-4)-	2.7308e-1(1.98e-3)-	2.7528e-1(3.17e-3)-	2.2395e-1(7.80e-2)-	2.7386e-1(1.64e-3)-	2.8121e-1(1.20e-3)
		10	1.0367e-2(1.39e-4)-	1.0103e-2(5.62e-4)-	1.0701e-2(1.11e-4)-	1.1126e-2(8.44e-5)+	1.0983e-2(1.01e-4)+	8.7785e-3(1.69e-3)-	1.0017e-2(1.21e-4)-	1.0795e-2(9.90e-5)
		15	4.4720e-4(2.55e-5)-	4.9750e-4(3.74e-5)-	5.1218e-4(2.08e-5)-	5.8810e-4(1.91e-5)+	5.0056e-4(8.76e-5)=	1.9812e-4(8.94e-5)-	3.1874e-4(3.37e-5)-	5.2202e-4(1.89e-5)
Disconnected	WFG2	3	9.2897e-1(1.57e-3)-	9.2498e-1(4.04e-3)-	9.3424e-1(1.12e-3)-	9.0996e-1(7.50e-3)-	9.3289e-1(1.10e-3)-	9.3537e-1(9.26e-4)-	9.2621e-1(1.93e-3)-	9.3669e-1(1.30e-3)
		10	9.9680e-1(8.68e-4)-	9.8735e-1(2.97e-3)-	9.9592e-1(8.18e-4)-	9.9056e-1(5.85e-3)-	9.9960e-1(2.65e-4)+	9.9514e-1(2.12e-3)-	9.8884e-1(2.27e-3)-	9.9790e-1(7.70e-4)
		15	9.9881e-1(6.69e-4)+	9.7432e-1(6.97e-3)-	9.9584e-1(1.05e-3)-	9.9432e-1(2.47e-3)-	9.9947e-1(3.89e-4)+	9.9378e-1(6.08e-3)-	9.6348e-1(6.01e-3)-	9.9818e-1(6.98e-4)
	MaF7	3	2.7357e-1(8.45e-4)-	2.7658e-1(6.94e-3)-	2.6704e-1(2.03e-2)-	2.7143e-1(1.70e-3)-	2.7008e-1(1.45e-2)-	2.6189e-1(3.56e-2)-	2.5191e-1(4.31e-3)-	2.7809e-1(9.45e-3)
		10	1.2505e-7(5.75e-7)-	1.9049e-1(5.16e-3)+	1.4301e-1(1.10e-2)+	1.4071e-1(5.15e-3)+	1.1077e-1(1.81e-2)-	1.7256e-1(3.71e-2)+	1.2411e-1(1.57e-2)=	1.2682e-1(1.05e-2)
		15	1.0351e-15(4.19e-15)-	1.3432e-1(1.63e-2)+	6.8747e-2(2.15e-2)+	7.7172e-2(3.73e-3)+	1.7401e-3(6.51e-3)-	1.3082e-1(1.30e-2)+	9.3519e-2(2.55e-2)+	2.9946e-2(3.58e-2)
	MaF11	3	9.2719e-1(1.54e-3)-	9.2355e-1(3.37e-3)-	9.3198e-1(1.63e-3)-	9.1208e-1(4.58e-3)-	9.3030e-1(1.50e-3)-	9.3290e-1(1.67e-3)-	9.2128e-1(2.35e-3)-	9.3473e-1(1.38e-3)
		10	9.8915e-1(2.29e-3)-	9.8307e-1(3.55e-3)-	9.9169e-1(2.03e-3)-	9.8599e-1(6.27e-3)-	9.9705e-1(1.00e-3)+	9.8773e-1(2.69e-3)-	9.8121e-1(5.38e-3)-	9.9522e-1(9.61e-4)
		15	9.9860e-1(7.96e-4)+	9.7296e-1(1.00e-2)-	9.9522e-1(9.91e-4)-	9.9266e-1(2.67e-3)-	9.9907e-1(7.68e-4)+	9.9541e-1(2.22e-3)-	9.6725e-1(1.05e-2)-	9.9772e-1(7.28e-4)
	DTLZ7	3	2.7341e-1(1.76e-3)-	2.7681e-1(6.85e-3)+	2.7836e-1(6.71e-3)+	2.7083e-1(1.36e-3)-	2.6542e-1(1.97e-2)-	2.5742e-1(2.98e-2)-	2.5289e-1(4.03e-3)-	2.7672e-1(1.13e-2)
		10	3.1140e-6(1.15e-5)-	1.9141e-1(1.08e-2)+	1.4939e-1(1.06e-2)+	1.4245e-1(4.25e-3)+	1.2218e-1(2.12e-2)-	1.9081e-1(7.39e-3)+	8.1606e-2(2.83e-2)-	1.3033e-1(1.43e-2)
		15	0.0000e+0(0.00e+0)-	1.4164e-1(5.12e-3)+	7.0911e-2(1.99e-2)+	8.4613e-2(2.55e-3)+	4.7869e-3(1.11e-2)-	1.3934e-1(1.13e-2)+	1.0437e-1(6.94e-3)+	3.0492e-2(3.37e-2)
Degenerated	WFG3	3	3.6374e-1(7.07e-3)-	3.8454e-1(5.64e-3)-	3.9561e-1(2.42e-3)+	3.5350e-1(1.09e-2)-	3.8141e-1(4.58e-3)-	4.0376e-1(2.09e-3)+	3.4822e-1(9.00e-3)-	3.8970e-1(5.07e-3)
		10	3.1675e-3(7.34e-3)=	0.0000e+0(0.00e+0)-	1.0620e-4(4.31e-4)-	3.4273e-2(1.96e-2)+	5.7262e-2(1.23e-2)+	6.1885e-2(1.59e-2)+	0.0000e+0(0.00e+0)-	1.3302e-3(3.86e-3)
		15	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	2.5509e-2(2.81e-2)+	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)
	MaF6	3	1.9958e-1(1.31e-4)-	1.9968e-1(8.81e-4)-	1.9995e-1(1.06e-4)-	1.8546e-1(2.00e-3)-	1.9979e-1(5.21e-4)-	1.8298e-1(1.58e-3)-	1.6088e-1(4.82e-2)-	2.0004e-1(1.52e-4)
		10	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	9.7472e-2(1.84e-2)+	2.2777e-3(7.48e-3)-	1.0066e-2(3.07e-2)=	1.8860e-2(3.84e-2)=	9.0421e-2(3.81e-3)+	6.7226e-3(2.56e-2)
		15	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	9.1970e-2(1.74e-2)+	8.7487e-2(2.33e-3)+	4.5619e-3(1.48e-2)=	6.8990e-3(1.82e-2)+	9.4892e-2(1.20e-3)+	4.0151e-4(2.20e-3)
	DTLZ6	3	2.0001e-1(1.22e-4)-	1.9978e-1(4.88e-4)-	2.0018e-1(6.66e-5)-	1.8045e-1(6.62e-3)-	2.0010e-1(5.58e-5)-	1.8393e-1(2.26e-4)-	1.6015e-1(2.56e-2)-	2.0035e-1(5.42e-5)
		10	0.0000e+0(0.00e+0)-	2.6539e-2(3.89e-2)=	8.1588e-2(1.40e-2)=	9.1790e-2(8.08e-4)+	2.2089e-7(1.21e-6)-	9.3390e-2(1.60e-3)+	8.3675e-2(2.35e-2)=	5.1355e-2(4.39e-2)
		15	0.0000e+0(0.00e+0)-	7.6950e-2(3.13e-2)+	8.2984e-2(2.29e-2)+	9.0889e-2(2.32e-4)+	1.1156e-2(2.79e-2)+	9.1588e-2(7.13e-4)+	9.0907e-2(2.16e-4)+	6.1303e-3(2.30e-2)
Regular	WFG1	3	9.3741e-1(4.80e-3)=	9.2723e-1(4.51e-3)-	9.3944e-1(3.99e-3)=	9.1590e-1(7.75e-3)-	9.4426e-1(2.06e-3)+	9.4136e-1(1.49e-3)+	9.2122e-1(1.48e-2)-	9.3985e-1(2.50e-3)
		10	9.9991e-1(5.82e-5)+	9.9420e-1(1.66e-3)-	9.9813e-1(4.14e-4)-	9.9083e-1(3.80e-2)-	9.9999e-1(3.14e-5)+	9.9745e-1(9.19e-4)-	9.9902e-1(2.34e-4)+	9.9838e-1(4.01e-4)
		15	9.9951e-1(5.08e-4)+	9.9221e-1(2.72e-3)-	9.9712e-1(7.45e-4)-	9.9829e-1(5.36e-4)=	9.9999e-1(8.66e-6)+	9.9814e-1(1.46e-3)=	9.9909e-1(2.13e-4)+	9.9830e-1(6.88e-4)
	WFG7	3	5.3639e-1(2.78e-3)-	5.5252e-1(2.64e-3)-	5.5915e-1(1.06e-3)-	5.4083e-1(3.63e-3)-	5.5792e-1(1.22e-3)-	5.5840e-1(6.62e-4)-	5.4712e-1(1.59e-3)-	5.6001e-1(1.05e-3)
		10	7.5344e-1(3.04e-2)-	8.3754e-1(3.18e-2)-	9.3797e-1(2.96e-3)-	9.2571e-1(5.30e-3)-	9.6152e-1(1.10e-3)+	9.6167e-1(8.81e-4)+	9.4179e-1(4.93e-3)-	9.5130e-1(3.12e-3)
		15	7.5040e-1(3.43e-2)-	7.5523e-1(6.61e-2)-	9.5498e-1(2.90e-3)-	9.3499e-1(2.23e-2)-	9.8348e-1(9.96e-4)+	9.9021e-1(1.30e-4)+	9.6276e-1(2.68e-2)=	9.7354e-1(5.98e-3)
	MaF5	3	5.3636e-1(5.17e-2)+	4.7746e-1(1.01e-1)-	5.1498e-1(1.08e-1)-	5.3631e-1(3.31e-3)+	4.4720e-1(1.54e-1)-	4.3356e-1(1.55e-1)-	5.3856e-1(5.38e-2)+	5.2470e-1(8.14e-2)
		10	2.1531e-3(6.01e-3)-	9.5903e-1(1.80e-3)-	9.3447e-1(6.02e-3)-	7.4988e-1(2.23e-2)-	5.2465e-1(1.55e-1)-	9.6574e-1(7.81e-3)+	8.9476e-1(2.09e-2)-	9.6572e-1(1.53e-3)
		15	2.4801e-3(9.94e-3)-	9.8330e-1(1.13e-3)=	9.5473e-1(6.89e-3)-	7.9451e-1(1.20e-1)-	5.2561e-1(4.01e-1)-	9.8546e-1(6.13e-3)+	8.9620e-1(3.74e-2)-	9.8268e-1(1.87e-3)
Outlier	MaF3	3	9.5869e-1(3.34e-3)=	9.3787e-1(3.90e-2)-	9.6092e-1(1.72e-3)+	9.4253e-1(1.06e-2)-	9.5230e-1(1.61e-2)=	9.3669e-1(4.59e-2)-	9.5109e-1(7.42e-3)-	9.5896e-1(2.12e-3)
		10	0.0000e+0(0.00e+0)-	4.2296e-1(4.33e-1)-	9.9978e-1(2.60e-4)+	9.7071e-1(1.46e-1)-	0.0000e+0(0.00e+0)-	9.9821e-1(1.80e-3)=	8.7300e-1(2.85e-1)-	9.9859e-1(8.80e-4)
		15	0.0000e+0(0.00e+0)-	3.7203e-1(4.50e-1)-	9.9901e-1(5.62e-4)+	9.9663e-1(2.89e-3)+	0.0000e+0(0.00e+0)-	9.7630e-1(5.20e-2)-	9.9763e-1(1.34e-3)=	9.9524e-1(1.03e-2)
	MaF9	3	6.8467e-1(5.02e-2)-	6.9024e-1(8.71e-2)-	8.3621e-1(1.01e-3)=	8.2458e-1(5.88e-3)-	6.4156e-1(1.04e-1)-	8.3648e-1(7.68e-3)+	8.3802e-1(2.34e-3)+	8.3594e-1(5.15e-3)
		10	2.7137e-5(7.42e-5)-	8.9627e-3(2.07e-3)+	1.4535e-2(1.29e-3)+	1.5798e-2(1.93e-4)+	2.6257e-3(1.42e-3)-	1.3472e-2(2.93e-3)+	4.3641e-3(8.53e-4)=	4.0275e-3(1.47e-3)
		15	2.4037e-4(1.87e-4)-	6.7398e-4(1.43e-4)-	1.1109e-3(2.44e-5)-	8.7027e-4(3.84e-5)-	5.8304e-4(2.52e-4)-	5.3486e-4(3.24e-4)-	7.2719e-4(3.06e-4)-	1.1304e-3(3.41e-5)
	DTLZ3	3	4.2854e-1(1.35e-1)=	4.0587e-1(1.73e-1)-	4.0835e-1(2.10e-1)=	3.4307e-1(2.30e-1)=	3.4344e-1(2.14e-1)=	4.6285e-1(1.34e-1)=	8.3754e-2(1.48e-1)-	4.1774e-1(1.98e-1)
		10	0.0000e+0(0.00e+0)-	5.5413e-1(2.66e-1)-	9.3455e-1(6.97e-2)-	7.8179e-1(3.56e-1)-	0.0000e+0(0.00e+0)-	9.6153e-1(4.84e-3)+	7.5293e-1(3.18e-1)-	9.5483e-1(5.89e-3)
		15	0.0000e+0(0.00e+0)-	6.1448e-1(1.66e-1)-	2.9675e-1(1.13e-1)-	8.6691e-1(2.22e-1)+	0.0000e+0(0.00e+0)-	9.8898e-1(5.60e-3)+	9.8833e-1(3.39e-3)+	8.5476e-1(1.65e-1)
Other	VNT1	3	3.4407e-1(3.80e-4)=	3.4226e-1(1.24e-3)-	3.4530e-1(4.43e-4)+	3.3895e-1(1.09e-3)-	3.4429e-1(5.39e-4)+	3.3917e-1(9.30e-4)-	3.4077e-1(1.30e-3)-	3.4379e-1(8.92e-4)
	VNT2	3	3.3312e-1(7.42e-5)-	3.3322e-1(2.70e-4)=	3.3369e-1(1.65e-4)+	3.2267e-1(8.09e-3)-	3.3345e-1(3.39e-4)+	3.3790e-1(3.76e-4)+	3.2929e-1(2.73e-3)-	3.3319e-1(1.88e-4)
	VNT3	3	1.7573e-1(4.91e-5)-	1.7605e-1(4.31e-5)+	1.7596e-1(8.32e-5)-	1.7264e-1(7.05e-4)-	1.7590e-1(4.24e-5)-	1.7543e-1(1.00e-4)-	1.7502e-1(3.61e-4)-	1.7601e-1(5.95e-5)
	IMOP4	3	4.3307e-1(3.34e-4)-	4.2658e-1(9.41e-3)-	4.3343e-1(6.91e-4)-	4.3073e-1(1.31e-3)-	4.3274e-1(3.89e-4)-	3.5053e-1(1.34e-1)-	4.2616e-1(2.46e-3)-	4.3408e-1(4.12e-4)
	IMOP5	3	4.8908e-1(3.59e-3)-	5.1744e-1(9.19e-3)+	5.0824e-1(4.92e-3)=	5.5396e-1(3.23e-2)+	5.1161e-1(1.14e-2)=	5.1777e-1(1.16e-2)+	4.8137e-1(4.34e-3)-	5.0708e-1(4.92e-3)
	IMOP6	3	5.2492e-1(7.77e-4)+	5.2586e-1(1.14e-3)+	5.2702e-1(1.87e-2)+	5.2412e-1(1.21e-3)+	5.2831e-1(6.16e-4)+	4.5743e-1(9.43e-2)-	2.3907e-1(1.83e-1)-	5.0592e-1(6.92e-2)
	IMOP7	3	5.0615e-1(5.50e-2)+	4.9351e-1(1.39e-2)+	4.4496e-1(9.49e-2)+	5.1627e-1(2.18e-3)+	5.2762e-1(7.77e-4)+	9.1550e-2(1.57e-3)-	5.1166e-1(2.06e-2)+	2.1004e-1(1.94e-1)
	IMOP8	3	5.0733e-1(2.89e-3)-	5.3344e-1(1.10e-2)+	5.2592e-1(3.36e-2)-	5.2198e-1(3.30e-3)-	5.4632e-1(1.89e-2)=	5.1842e-1(4.75e-2)-	4.8361e-1(1.80e-2)-	5.2669e-1(3.85e-2)
+/ − / ≈			7/44/14	21/38/6	27/31/7	24/35/6	20/34/11	30/30/5	16/42/7	

Table A.10
HV values obtained by TS-MOEA-EPF and 6 of 13 MaOEAs, where "+", "-", and "=" respectively represent the comparison algorithms being superior, inferior, and equivalent to TS-MOEA-EPF (Part II)

Property	Problem	M	AdaW	MultiGPO	TS-NSGA-II	AR-MOEA	hpaEA	CARV-MOEA	TS-MOEA-EPF
Inverted	IDTLZ1	3	2.1996e-1(5.85e-3)=	2.1973e-1(2.48e-3)-	1.9941e-1(4.92e-3)-	2.1600e-1(4.52e-3)-	2.1854e-1(3.06e-3)-	2.1833e-1(3.09e-3)-	2.2127e-1(3.18e-3)
		10	4.0937e-7(5.29e-7)=	3.9156e-7(6.12e-7)=	9.6618e-8(5.17e-8)-	2.6647e-7(5.10e-7)=	5.0703e-7(5.24e-7)=	6.2078e-7(4.98e-7)+	3.5512e-7(5.43e-7)
		15	0.0000e+0(0.00e+0)=	3.2659e-13(5.51e-13)+	4.8570e-13(2.72e-13)+	1.7696e-13(6.02e-13)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)
	IDTLZ2	3	5.3964e-1(5.12e-4)-	5.3199e-1(1.21e-3)-	5.1758e-1(1.20e-3)-	5.2930e-1(8.20e-4)-	5.3078e-1(1.92e-3)-	5.3879e-1(8.29e-4)-	5.4184e-1(4.47e-4)
		10	5.1895e-5(1.11e-5)-	1.0543e-4(1.25e-5)+	1.6836e-4(3.27e-6)+	2.4221e-4(2.17e-5)+	1.3860e-4(9.85e-6)+	3.6698e-4(2.52e-5)+	9.1574e-5(1.97e-5)
		15	1.3188e-9(7.22e-9)=	1.4508e-8(1.91e-8)+	1.7107e-7(1.11e-8)+	1.7231e-7(5.12e-8)+	1.4410e-7(8.02e-8)+	2.4317e-7(7.56e-8)+	0.0000e+0(0.00e+0)
	MaF1	3	2.2370e-1(1.90e-4)-	2.2204e-1(4.07e-4)-	2.0491e-1(7.53e-4)-	2.2174e-1(6.48e-4)-	2.2027e-1(6.41e-4)-	2.2129e-1(4.58e-4)-	2.2396e-1(1.88e-4)
		10	4.6474e-7(6.29e-7)=	4.9056e-7(6.22e-7)=	2.5404e-7(9.19e-8)+	5.9996e-7(7.70e-7)+	5.1247e-7(5.85e-7)=	6.0549e-7(4.56e-7)+	2.3331e-7(4.30e-7)
		15	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	1.0682e-12(5.61e-13)+	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)
	MaF4	3	5.3166e-1(5.43e-3)=	5.1417e-1(4.96e-3)-	4.8876e-1(2.52e-2)-	5.2124e-1(5.84e-3)-	5.2213e-1(5.54e-3)-	5.2446e-1(3.74e-3)-	5.3186e-1(5.82e-3)
		10	4.9731e-5(2.19e-5)-	6.8092e-5(1.20e-5)-	1.7377e-4(2.47e-5)+	3.5609e-6(1.67e-6)-	1.0085e-4(2.00e-5)+	4.2970e-6(1.32e-6)-	8.6946e-5(2.02e-5)
		15	2.3996e-9(1.31e-8)=	2.9160e-9(9.08e-9)=	9.6719e-8(6.48e-8)+	9.8591e-12(5.29e-12)+	8.7092e-8(9.01e-8)+	7.7588e-12(2.43e-12)+	0.0000e+0(0.00e+0)
Partial	MaF2	3	2.4513e-1(6.24e-4)-	2.4412e-1(8.41e-4)-	2.4041e-1(1.44e-3)-	2.4182e-1(9.74e-4)-	2.4251e-1(1.08e-3)-	2.4410e-1(8.18e-4)-	2.4618e-1(3.22e-4)
		10	1.8491e-1(6.16e-3)-	2.1324e-1(4.28e-3)+	2.0379e-1(2.08e-3)=	2.0332e-1(3.18e-3)=	2.2008e-1(3.03e-3)+	2.2240e-1(2.04e-3)+	2.0198e-1(6.25e-3)
		15	1.1561e-1(9.87e-3)-	1.5341e-1(4.04e-3)=	1.5292e-1(3.19e-3)=	1.7772e-1(5.63e-3)+	1.7963e-1(3.96e-3)+	2.0132e-1(3.35e-3)+	1.5555e-1(9.68e-3)
	MaF8	3	2.7584e-1(5.14e-3)-	2.7860e-1(1.03e-3)-	2.5057e-1(3.12e-3)-	2.7463e-1(9.12e-4)-	2.5407e-1(4.31e-2)-	2.7728e-1(3.13e-3)-	2.8121e-1(1.20e-3)
		10	1.0396e-2(2.64e-4)-	1.1085e-2(1.13e-4)+	8.1028e-3(4.14e-4)-	1.0831e-2(1.38e-4)=	1.1062e-2(1.16e-4)+	1.0603e-2(2.90e-4)-	1.0795e-2(9.90e-5)
		15	4.8204e-4(1.88e-5)-	5.6722e-4(1.82e-5)+	1.4295e-4(6.36e-5)-	5.3973e-4(1.76e-5)+	5.8339e-4(3.50e-5)+	5.1364e-4(2.21e-5)=	5.2202e-4(1.89e-5)
Disconnected	WFG2	3	9.3455e-1(7.19e-4)-	9.3260e-1(1.38e-3)-	9.2951e-1(1.49e-3)-	9.3286e-1(8.09e-4)-	9.3023e-1(2.21e-3)-	9.2261e-1(5.56e-3)-	9.3669e-1(1.30e-3)
		10	9.9894e-1(4.68e-4)+	9.9454e-1(9.35e-4)-	9.9708e-1(9.17e-4)-	9.9844e-1(6.76e-4)+	9.9377e-1(1.13e-3)-	9.6199e-1(7.49e-3)-	9.9790e-1(7.70e-4)
		15	9.9752e-1(1.19e-3)-	9.9547e-1(2.40e-3)-	9.9712e-1(1.51e-3)-	9.9855e-1(7.19e-4)+	9.7957e-1(3.22e-2)-	9.5415e-1(1.16e-2)-	9.9818e-1(6.98e-4)
	MaF7	3	2.7999e-1(4.29e-4)+	2.7151e-1(1.47e-2)-	2.6959e-1(1.10e-2)-	2.6689e-1(1.97e-2)-	2.7119e-1(6.99e-3)-	2.7812e-1(9.96e-4)+	2.7809e-1(9.45e-3)
		10	4.3294e-2(2.08e-2)-	1.1739e-1(2.22e-2)=	1.7770e-1(5.49e-3)+	1.2214e-1(9.82e-3)=	1.2821e-1(1.26e-2)=	1.9045e-2(8.76e-3)-	1.2682e-1(1.05e-2)
		15	4.8235e-3(1.82e-2)-	8.6506e-2(1.42e-2)+	1.4688e-1(7.06e-3)+	4.7314e-2(2.39e-2)=	1.5215e-1(6.09e-3)+	2.6625e-5(7.96e-5)-	2.9946e-2(3.58e-2)
	MaF11	3	9.3230e-1(1.48e-3)-	9.3168e-1(1.66e-3)-	9.2654e-1(1.43e-3)-	9.3036e-1(1.44e-3)-	9.2498e-1(4.89e-3)-	9.1908e-1(4.68e-3)-	9.3473e-1(1.38e-3)
		10	9.9237e-1(2.68e-3)-	9.9061e-1(1.97e-3)-	9.9253e-1(2.04e-3)-	9.8985e-1(2.02e-3)-	9.8412e-1(2.82e-3)-	9.3831e-1(8.12e-3)-	9.9522e-1(9.61e-4)
		15	9.9606e-1(1.80e-3)-	9.9490e-1(2.22e-3)-	9.9600e-1(1.78e-3)-	9.9730e-1(1.06e-3)=	9.8287e-1(6.08e-3)-	9.4810e-1(1.03e-2)-	9.9772e-1(7.28e-4)
	DTLZ7	3	2.8003e-1(5.30e-4)=	2.7398e-1(1.26e-2)-	2.7173e-1(8.07e-3)-	2.6666e-1(1.95e-2)-	2.7019e-1(1.32e-2)-	2.7823e-1(1.31e-3)+	2.7672e-1(1.13e-2)
		10	6.4976e-2(1.87e-2)-	1.1483e-1(2.15e-2)-	1.7809e-1(4.87e-3)+	1.3388e-1(8.70e-3)=	1.5957e-1(6.55e-3)+	1.4342e-2(8.46e-3)-	1.3033e-1(1.43e-2)
		15	1.0239e-1(3.70e-2)+	8.3617e-2(1.31e-2)+	1.3114e-1(9.03e-3)+	5.4102e-2(1.85e-2)+	1.5364e-1(3.78e-3)+	1.7050e-4(4.65e-4)-	3.0492e-2(3.37e-2)
Degenerated	WFG3	3	9.3273e-1(3.60e-3)+	4.0587e-1(2.44e-3)+	3.8036e-1(7.84e-3)-	3.7866e-1(3.83e-3)-	3.7467e-1(1.22e-2)-	3.7252e-1(5.21e-3)-	3.8970e-1(5.07e-3)
		10	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	3.0423e-2(1.92e-2)+	9.9582e-4(3.95e-3)=	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	1.3302e-3(3.86e-3)
		15	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)
	MaF6	3	1.9986e-1(2.62e-4)-	1.9857e-1(3.78e-4)-	1.8855e-1(2.49e-3)-	1.9958e-1(1.73e-4)-	1.9976e-1(1.39e-4)-	1.9964e-1(4.77e-4)-	2.0004e-1(1.52e-4)
		10	1.0075e-1(3.97e-4)+	6.0006e-2(4.98e-2)+	9.6931e-2(2.92e-3)+	6.3789e-2(4.94e-2)+	6.7493e-3(2.57e-2)=	5.9409e-2(4.89e-2)+	6.7226e-3(2.56e-2)
		15	9.5355e-2(2.43e-4)+	3.1336e-2(2.72e-2)+	8.3907e-2(4.99e-3)+	7.4668e-2(2.70e-2)+	0.0000e+0(0.00e+0)=	5.1703e-2(4.46e-2)+	4.0151e-4(2.20e-3)
	DTLZ6	3	2.0026e-1(3.75e-5)-	1.9915e-1(3.02e-4)-	1.8569e-1(2.54e-3)-	1.9994e-1(7.30e-5)-	2.0015e-1(4.61e-5)-	1.9321e-1(3.65e-2)-	2.0035e-1(5.42e-5)
		10	0.0000e+0(0.00e+0)-	8.9520e-2(6.02e-3)+	8.4183e-2(2.70e-2)+	9.1118e-2(9.31e-4)+	0.0000e+0(0.00e+0)-	8.9126e-2(1.01e-2)+	5.1355e-2(4.39e-2)
		15	1.3563e-2(3.10e-2)=	9.0880e-2(2.36e-4)+	8.8857e-2(1.26e-2)+	9.1297e-2(5.89e-4)+	0.0000e+0(0.00e+0)-	9.0949e-2(4.52e-4)+	6.1303e-3(2.30e-2)
Regular	WFG1	3	9.3774e-1(5.41e-3)=	9.3140e-1(2.57e-3)-	9.3738e-1(4.07e-3)-	9.4305e-1(2.41e-3)+	6.6965e-1(1.43e-1)-	9.0567e-1(8.59e-3)-	9.3985e-1(2.50e-3)
		10	9.9965e-1(2.91e-4)+	9.9857e-1(6.75e-4)=	9.9995e-1(3.99e-5)+	9.9994e-1(4.07e-5)+	9.9098e-1(3.00e-2)-	9.9625e-1(8.25e-4)-	9.9838e-1(4.01e-4)
		15	9.9995e-1(6.50e-5)+	9.9800e-1(9.38e-4)=	9.9999e-1(1.10e-5)+	9.9982e-1(1.72e-4)+	9.9510e-1(7.76e-3)-	9.9586e-1(8.14e-4)-	9.9830e-1(6.88e-4)
	WFG7	3	5.5670e-1(1.16e-3)-	5.4434e-1(3.14e-3)-	5.5627e-1(8.07e-4)-	5.5495e-1(1.07e-3)-	5.5617e-1(8.88e-4)-	5.4788e-1(2.22e-3)-	5.6001e-1(1.05e-3)
		10	7.8330e-1(2.85e-2)-	9.6459e-1(3.29e-3)+	9.6415e-1(9.20e-4)+	9.4870e-1(2.48e-3)-	8.4765e-1(2.26e-2)-	4.3425e-1(4.03e-2)-	9.5130e-1(3.12e-3)
		15	7.5777e-1(3.70e-2)-	9.6696e-1(8.54e-3)-	9.8796e-1(1.25e-3)+	9.8494e-1(1.71e-3)+	7.2900e-1(2.98e-2)-	3.1212e-1(4.54e-2)-	9.7354e-1(5.98e-3)
	MaF5	3	5.3798e-1(6.57e-2)+	4.9731e-1(5.27e-2)-	5.5772e-1(7.16e-4)+	4.3620e-1(1.25e-1)-	4.2262e-1(1.51e-1)-	5.2896e-1(6.19e-2)+	5.2470e-1(8.14e-2)
		10	1.1025e-1(5.72e-2)-	8.3009e-1(6.85e-3)-	9.5564e-1(4.35e-3)-	8.9705e-1(1.51e-2)-	6.6922e-1(6.27e-2)-	5.5980e-1(4.02e-2)-	9.6572e-1(1.53e-3)
		15	0.0000e+0(0.00e+0)-	8.7163e-1(1.65e-1)-	9.8428e-1(2.41e-3)+	9.7396e-1(6.38e-3)-	8.9699e-1(3.21e-2)-	4.2018e-1(3.80e-2)-	9.8268e-1(1.87e-3)
Outlier	MaF3	3	9.5970e-1(2.74e-3)=	9.4681e-1(3.52e-3)-	8.8384e-1(7.27e-2)-	9.5529e-1(2.63e-3)-	9.5768e-1(2.30e-3)-	9.4653e-1(4.35e-3)-	9.5896e-1(2.12e-3)
		10	0.0000e+0(0.00e+0)-	9.9962e-1(3.97e-4)+	7.9529e-1(1.26e-1)-	9.6656e-1(1.83e-1)-	0.0000e+0(0.00e+0)-	9.9247e-1(2.55e-2)-	9.9859e-1(8.80e-4)
		15	5.4400e-2(2.07e-1)-	9.4594e-1(2.08e-1)-	5.2117e-1(4.43e-1)-	9.9952e-1(9.26e-4)+	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	9.9524e-1(1.03e-2)
	MaF9	3	6.2447e-1(2.79e-1)-	8.3343e-1(1.65e-3)-	6.9429e-1(8.27e-2)-	8.3807e-1(1.52e-3)+	8.3021e-1(1.88e-2)=	8.2481e-1(8.53e-3)-	8.3594e-1(5.15e-3)
		10	7.0932e-4(8.11e-4)-	1.8751e-2(1.35e-4)+	9.9767e-3(2.48e-3)+	1.6527e-2(2.42e-4)+	1.3238e-2(4.55e-3)+	9.9265e-3(6.95e-4)+	4.0275e-3(1.47e-3)
		15	1.2167e-4(1.50e-4)-	1.0950e-3(2.09e-4)=	4.2554e-4(2.09e-4)-	1.0811e-3(2.82e-5)-	1.1245e-3(4.21e-5)=	9.4266e-4(7.69e-5)-	1.1304e-3(3.41e-5)
	DTLZ3	3	3.1861e-1(1.97e-1)-	4.1094e-1(2.12e-1)=	3.7731e-1(2.09e-1)=	4.4639e-1(1.43e-1)=	3.9472e-1(1.96e-1)=	3.2865e-1(1.99e-1)-	4.1774e-1(1.98e-1)
		10	0.0000e+0(0.00e+0)-	8.6931e-1(2.30e-1)-	8.9032e-1(2.92e-2)-	9.4489e-1(1.55e-2)-	0.0000e+0(0.00e+0)-	9.4700e-1(7.99e-3)-	9.5483e-1(5.89e-3)
		15	1.1385e-2(6.24e-2)-	7.9966e-1(2.24e-1)-	3.0186e-1(1.29e-1)-	9.7714e-1(6.21e-3)+	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	8.5476e-1(1.65e-1)
Other	VNT1	3	3.4537e-1(3.18e-4)+	3.3895e-1(1.29e-3)-	3.3960e-1(9.67e-4)-	3.4032e-1(1.34e-3)-	3.4085e-1(5.75e-4)-	3.4025e-1(1.92e-3)-	3.4379e-1(8.92e-4)
	VNT2	3	3.3334e-1(4.75e-5)+	3.3094e-1(1.97e-3)-	3.3131e-1(5.66e-4)-	3.3183e-1(6.12e-4)-	3.3587e-1(3.51e-3)+	3.3408e-1(1.61e-3)=	3.3319e-1(1.88e-4)
	VNT3	3	1.7595e-1(1.93e-5)-	1.7556e-1(1.28e-4)-	1.7431e-1(3.50e-4)-	1.7609e-1(1.95e-5)+	1.7600e-1(4.35e-5)=	1.7615e-1(4.90e-5)+	1.7601e-1(5.95e-5)
	IMOP4	3	4.2864e-1(1.87e-2)-	4.3204e-1(1.05e-3)-	4.1899e-1(3.60e-3)-	4.3424e-1(2.18e-4)=	3.1694e-2(2.71e-3)-	4.3095e-1(2.04e-3)-	4.3408e-1(4.12e-4)
	IMOP5	3	5.1869e-1(1.34e-2)+	5.0247e-1(3.42e-3)-	5.0791e-1(8.32e-3)=	5.0822e-1(3.76e-3)=	4.2699e-1(4.13e-2)-	5.0939e-1(4.70e-3)=	5.0708e-1(4.92e-3)
	IMOP6	3	5.3177e-1(1.14e-3)+	5.3074e-1(6.02e-4)+	5.2252e-1(6.45e-4)+	5.1054e-1(5.92e-2)+	2.9443e-1(8.95e-2)-	5.2478e-1(2.00e-2)+	5.0592e-1(6.92e-2)
	IMOP7	3	4.9283e-1(8.28e-2)+	1.3750e-1(1.31e-1)-	4.8423e-1(7.99e-2)+	1.5303e-1(1.50e-1)-	9.1019e-2(3.64e-4)-	9.1946e-2(1.67e-3)-	2.1004e-1(1.94e-1)
	IMOP8	3	5.3052e-1(3.06e-3)+	5.2421e-1(3.90e-2)-	5.2564e-1(6.40e-3)-	5.0474e-1(2.22e-2)-	4.3428e-1(9.69e-2)-	5.2502e-1(2.18e-3)-	5.2669e-1(3.85e-2)
+/ − / ≈			15/37/13	17/37/11	25/35/5	23/28/14	13/40/12	17/42/6	

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Table A.11
IGD+ values obtained by TS-MOEA-EPF and 7 of 13 MaOEAs on ZCAT, where "+", "-", and "=" respectively represent the comparison algorithms being superior, inferior, and equivalent to TS-MOEA-EPF (Part I)

Property	Problem	M	CA-MOEA	DEA-GNG	RVEA-iGNG	PeEA	LMPFE	HEA	RVEA-AdaK	TS-MOEA-EPF
Regular	ZCAT1	2	1.0388e-2(1.00e-2)-	1.8726e-2(2.16e-2)-	1.8485e-2(1.47e-2)-	1.8865e-2(9.99e-3)-	2.1955e-2(1.41e-2)-	3.0735e-2(3.28e-2)-	6.0652e-2(2.10e-2)-	1.0305e-2(1.55e-2)
		3	1.3601e-1(2.80e-2)-	1.1158e-1(1.98e-2)-	9.4993e-2(2.54e-2)=	1.4466e-1(1.28e-2)-	1.2402e-1(2.96e-2)-	7.7520e-2(1.79e-1)+	1.4528e-1(2.86e-2)-	9.4516e-2(2.38e-2)
	ZCAT2	2	1.2558e-2(1.10e-2)=	1.4935e-2(1.43e-2)=	4.0328e-2(3.47e-2)-	2.1141e-2(1.37e-2)=	1.7215e-2(1.40e-2)=	4.2754e-2(2.99e-2)-	5.9637e-2(2.85e-2)-	2.5093e-2(2.29e-2)
		3	1.0046e-1(4.24e-2)-	1.0201e-1(3.81e-2)-	5.9577e-2(2.67e-2)=	1.0456e-1(3.25e-2)-	9.1870e-2(4.53e-2)-	1.2528e-1(6.76e-2)-	1.1756e-1(4.64e-2)-	6.8838e-2(3.55e-2)
Partial	ZCAT3	2	1.1031e-2(1.02e-2)+	1.7296e-2(1.74e-2)-	2.6304e-2(2.87e-2)-	1.8817e-2(9.75e-3)-	1.6675e-2(1.13e-2)-	2.5759e-2(2.20e-2)-	6.6257e-2(3.41e-2)-	1.1482e-2(1.26e-2)
		3	2.4958e-1(5.45e-2)-	2.8359e-1(2.54e-1)-	9.4576e-2(1.63e-2)=	1.2056e-1(1.40e-2)-	1.8901e-1(3.46e-2)-	1.1391e-1(1.91e-2)-	3.2026e-1(6.25e-2)-	1.0091e-1(2.38e-2)
	ZCAT4	2	1.4408e-2(3.80e-2)=	3.1769e-2(4.22e-2)+	2.3163e-1(1.78e-1)-	4.1218e-2(4.00e-2)-	6.0893e-2(5.81e-2)-	3.0041e-1(2.09e-1)-	4.9911e-2(5.54e-2)-	3.6389e-2(4.66e-2)
		3	2.3807e-1(3.46e-2)-	1.6356e-1(8.80e-2)-	2.1419e-1(1.17e-1)-	1.6025e-1(4.83e-2)-	2.1564e-1(6.40e-2)-	1.9099e-1(6.85e-2)-	3.1374e-1(8.06e-2)-	1.3160e-1(4.27e-2)
	ZCAT8	2	8.8993e-3(9.00e-3)+	1.6879e-2(2.29e-2)-	3.9463e-2(3.89e-2)-	2.9222e-2(2.16e-2)-	3.2373e-2(2.84e-2)-	3.8939e-2(2.66e-2)-	8.8011e-2(3.40e-2)-	1.3986e-2(1.94e-2)
		3	3.7243e-1(1.22e-1)-	4.2254e-1(2.73e-1)-	3.6268e-1(2.42e-1)-	2.6682e-1(9.70e-2)-	2.4981e-1(1.28e-1)-	9.6395e-2(3.04e-2)=	4.2441e-1(1.28e-1)-	1.0479e-1(4.17e-2)
	ZCAT9	2	1.4445e-2(1.33e-2)=	2.8406e-2(2.42e-2)-	5.0275e-2(5.88e-2)-	3.1244e-2(1.88e-2)-	2.1926e-2(1.74e-2)=	3.0748e-2(2.79e-2)-	8.2822e-2(2.92e-2)-	1.7409e-2(1.55e-2)
		3	1.9231e-1(4.69e-2)-	3.0361e-1(3.51e-1)-	1.2585e-1(1.62e-1)=	1.3411e-1(1.82e-2)-	1.7757e-1(7.63e-2)-	9.5613e-2(1.77e-2)=	2.9096e-1(4.20e-2)-	8.9216e-2(2.41e-2)
Flat	ZCAT5	2	1.4152e-2(1.69e-2)=	2.4756e-2(2.45e-2)=	4.7129e-2(4.37e-2)=	2.9767e-2(2.41e-2)=	1.4058e-2(1.64e-2)+	1.1936e-1(8.14e-2)-	6.2667e-2(3.28e-2)-	3.4794e-2(2.83e-2)
		3	1.9244e-1(2.89e-2)-	1.7617e-1(6.35e-2)-	9.1218e-2(1.41e-2)-	8.9915e-2(9.85e-3)-	1.8405e-1(2.63e-2)-	1.1262e-1(2.81e-2)-	1.8177e-1(3.53e-2)-	6.7062e-2(1.14e-2)
	ZCAT6	2	1.7086e-2(2.27e-2)=	2.9848e-2(3.67e-2)=	8.5057e-2(5.57e-2)-	3.9084e-2(3.55e-2)-	4.5259e-2(4.16e-2)=	1.0911e-1(6.22e-2)-	1.2243e-1(5.56e-2)-	2.4683e-2(3.48e-2)
		3	1.8392e-1(3.79e-2)-	1.0035e-1(1.91e-2)-	1.0271e-1(1.69e-2)-	9.2165e-2(9.96e-3)-	1.4641e-1(2.51e-2)-	8.9796e-2(1.70e-2)-	1.8748e-1(2.28e-2)-	7.6013e-2(1.22e-2)
	ZCAT7	2	2.3768e-2(6.08e-2)=	7.7252e-2(1.06e-1)=	2.0527e-1(1.61e-1)-	2.1568e-2(1.90e-2)=	5.5581e-2(8.56e-2)=	3.3646e-1(1.68e-1)-	9.7843e-2(7.96e-2)=	1.2668e-1(1.42e-1)
		3	1.9585e-1(2.82e-2)-	1.4117e-1(5.94e-2)-	1.6553e-1(8.58e-2)-	1.3772e-1(2.58e-2)-	1.6854e-1(3.62e-2)-	1.6595e-1(9.42e-2)-	2.6577e-1(4.89e-2)-	1.2447e-1(7.30e-2)
	ZCAT10	2	2.4919e-2(2.69e-2)+	3.3262e-2(3.32e-2)-	8.5536e-2(1.08e-1)-	6.6378e-2(3.75e-2)-	4.5879e-2(4.07e-2)-	3.4591e-1(3.38e-1)-	1.3384e-1(5.58e-2)-	2.7869e-2(2.99e-2)
		3	2.0313e-1(1.95e-2)-	1.2211e-1(2.83e-2)-	2.1564e-1(3.83e-2)-	2.3821e-1(8.45e-2)-	1.7406e-1(2.08e-2)-	1.6305e-1(1.21e-2)-	2.4759e-1(4.51e-2)-	8.2792e-2(1.43e-2)
Disconnected	ZCAT11	2	1.1637e-2(9.89e-3)=	2.9063e-2(5.89e-2)=	1.2382e-1(1.09e-1)-	1.8828e-2(8.25e-3)+	6.2161e-2(7.47e-2)-	1.5687e-1(1.72e-1)-	6.9479e-2(3.16e-2)-	3.4028e-2(6.42e-2)
		3	2.2837e-1(4.40e-2)-	2.5684e-1(2.09e-1)-	9.9008e-2(7.17e-2)=	1.1059e-1(1.49e-2)-	1.7589e-1(4.83e-2)-	9.4846e-2(2.58e-2)=	3.1251e-1(4.46e-2)-	8.9837e-2(1.87e-2)
	ZCAT12	2	1.0720e-2(1.03e-2)=	8.0778e-3(7.22e-3)=	1.1506e-1(1.80e-1)-	1.8586e-2(9.69e-3)=	1.9047e-2(1.88e-2)=	2.3673e-1(3.05e-1)-	6.7830e-2(4.54e-2)-	4.2620e-2(6.96e-2)
		3	2.8337e-1(5.06e-2)-	2.7841e-1(1.84e-1)-	1.1290e-1(2.96e-2)-	1.5632e-1(3.75e-2)-	1.8876e-1(5.27e-2)-	1.3638e-1(2.19e-2)-	4.7566e-1(8.74e-2)-	9.6159e-2(1.73e-2)
	ZCAT13	2	1.6015e-2(1.50e-2)=	1.7901e-2(1.51e-2)=	7.9115e-2(9.32e-2)-	1.8534e-2(1.30e-2)=	4.7654e-2(7.78e-2)=	1.2968e-1(9.85e-2)-	5.7506e-2(2.53e-2)-	5.7088e-2(8.54e-2)
		3	2.2645e-1(3.75e-2)-	2.1694e-1(1.37e-1)-	1.1510e-1(8.50e-2)=	1.1424e-1(1.99e-2)-	1.9833e-1(5.70e-2)-	9.6959e-2(2.78e-2)=	2.8109e-1(9.78e-2)-	9.2133e-2(3.05e-2)
Degenerated	ZCAT14	2	1.4449e-2(5.91e-2)-	6.8946e-2(1.34e-1)=	2.2054e-1(1.80e-1)-	5.3319e-3(2.90e-3)+	5.5665e-3(3.49e-3)=	3.2318e-2(2.90e-2)-	7.4217e-2(1.31e-1)-	1.1780e-2(1.37e-2)
		3	1.7191e-1(2.31e-2)-	2.2222e-1(1.35e-1)-	2.9445e-1(2.60e-1)-	1.2334e-1(1.98e-2)-	1.5012e-1(3.28e-2)-	7.5224e-2(2.74e-2)+	2.3296e-1(4.94e-2)-	1.0818e-1(3.50e-2)
Mixed	ZCAT15	2	9.7317e-3(8.04e-3)=	2.5096e-2(7.65e-2)=	5.1921e-2(8.72e-2)-	2.4795e-2(8.10e-3)-	3.3408e-2(8.00e-2)=	2.0344e-1(1.96e-1)-	5.0448e-2(1.46e-2)-	2.4324e-2(5.66e-2)
		3	7.0402e-1(2.17e-1)-	3.8761e-1(1.36e-1)-	3.4064e-1(1.48e-1)-	3.4075e-1(1.85e-1)-	3.5154e-1(9.64e-2)-	1.1105e-1(8.84e-2)+	5.8421e-1(1.63e-1)-	1.8745e-1(9.92e-2)
	ZCAT16	2	2.6053e-2(6.04e-2)=	3.4909e-2(5.15e-2)-	7.9324e-2(1.01e-1)-	2.0730e-2(1.13e-2)=	3.1522e-2(3.98e-2)-	1.6052e-2(1.54e-1)-	6.9383e-2(2.99e-2)-	3.0642e-2(5.54e-2)
		3	4.5121e-1(1.61e-1)-	2.4205e-1(6.13e-2)-	1.2399e-1(3.02e-2)-	1.2862e-1(2.62e-2)-	2.5554e-1(4.01e-2)-	7.9484e-2(1.60e-2)+	2.9687e-1(2.88e-2)-	9.8513e-2(2.60e-2)
	ZCAT17	2	1.2372e-2(2.29e-2)=	3.5741e-2(3.16e-2)=	5.5086e-2(4.33e-2)-	3.5147e-2(2.38e-2)-	2.2517e-2(2.37e-2)=	8.6639e-2(4.88e-2)-	9.0457e-2(3.60e-2)-	2.7871e-2(2.87e-2)
		3	2.0308e-1(3.55e-2)-	1.2087e-1(6.21e-2)-	9.1946e-2(2.94e-2)-	9.4401e-2(1.78e-2)-	1.6760e-1(1.76e-2)-	1.0604e-1(6.82e-2)-	1.6652e-1(3.34e-2)-	5.3801e-2(9.41e-3)
	ZCAT18	2	3.6033e-2(6.89e-2)+	9.8054e-2(1.15e-1)=	2.3211e-1(1.57e-1)-	2.1909e-2(1.78e-2)+	5.9263e-2(8.83e-2)=	3.5165e-1(8.37e-2)-	1.1777e-1(8.20e-2)=	1.5723e-1(1.50e-1)
		3	2.1291e-1(5.92e-2)-	1.3279e-1(4.46e-2)-	1.3691e-1(6.51e-2)-	1.5682e-1(1.14e-1)-	1.7485e-1(2.87e-2)-	1.3425e-1(7.45e-2)=	2.9301e-1(5.82e-2)-	9.4246e-2(4.08e-2)
	ZCAT19	2	1.5122e-2(1.49e-2)=	1.4097e-2(1.52e-2)=	3.6083e-2(2.28e-2)-	2.8098e-2(1.58e-2)-	2.3761e-2(1.76e-2)-	3.1765e-2(2.39e-2)=	5.7696e-2(2.94e-2)-	2.0734e-2(1.77e-2)
		3	2.9077e-1(5.20e-2)-	1.9307e-1(5.50e-2)-	1.1697e-1(2.49e-2)=	1.1547e-1(1.67e-2)=	2.4551e-1(3.47e-2)-	1.2199e-1(2.32e-2)=	3.2968e-1(5.14e-2)-	1.1411e-1(2.35e-2)
	ZCAT20	2	2.6305e-2(5.28e-2)=	4.3596e-2(8.48e-2)=	2.8825e-1(1.46e-1)-	3.2744e-2(2.80e-2)=	5.1249e-2(9.32e-2)=	2.7933e-1(1.69e-1)-	1.1450e-1(7.05e-2)=	1.3678e-1(1.58e-1)
		3	1.5117e-1(2.82e-2)=	1.4295e-1(8.73e-2)=	1.7860e-1(1.43e-1)=	1.1472e-1(3.27e-2)=	1.8601e-1(7.52e-2)=	2.5535e-1(1.90e-1)=	3.0959e-1(9.58e-2)-	1.9121e-1(1.60e-1)
+/ − / ≈			4/21/15	1/25/14	0/31/9	3/28/9	1/27/12	4/28/8	0/37/3	

Table A.12
IGD+ values obtained by TS-MOEA-EPF and 6 of 13 MaOEAs on ZCAT, where "+", "-", and "=" respectively represent the comparison algorithms being superior, inferior, and equivalent to TS-MOEA-EPF (Part II)

Property	Problem	M	AdaW	MultiGPO	TS-NSGA-II	AR-MOEA	hpaEA	CARV-MOEA	TS-MOEA-EPF
Regular	ZCAT1	2	1.1686e-2(1.26e-2)-	1.6658e-2(1.18e-2)-	6.1274e-2(4.47e-2)-	1.8757e-2(1.72e-2)-	2.5380e-2(2.97e-2)-	1.6411e-2(1.35e-2)-	1.0305e-2(1.55e-2)
		3	9.7433e-2(1.87e-2)=	1.1057e-1(1.19e-2)-	1.2128e-1(2.25e-2)-	6.1430e-3(7.14e-3)+	1.0113e-1(2.10e-2)=	1.4520e-2(6.17e-3)+	9.4516e-2(2.38e-2)
	ZCAT2	2	8.4690e-3(1.15e-2)+	2.2990e-2(2.05e-2)=	8.4804e-2(6.71e-2)-	2.8447e-2(2.14e-2)=	8.2516e-2(7.01e-2)-	2.6491e-2(1.84e-2)=	2.5093e-2(2.29e-2)
		3	7.2974e-2(3.69e-2)=	1.3833e-1(3.65e-2)-	1.4300e-1(7.09e-2)-	9.7751e-2(3.10e-2)-	1.1887e-1(3.49e-2)-	1.3979e-1(3.89e-2)-	6.8838e-2(3.55e-2)
Partial	ZCAT3	2	1.0562e-2(9.53e-3)=	1.9213e-2(1.27e-2)-	8.3105e-2(4.38e-2)-	2.0040e-2(1.53e-2)-	4.2772e-2(5.31e-2)-	2.0264e-2(1.30e-2)-	1.1482e-2(1.26e-2)
		3	1.4912e-1(2.74e-2)-	1.0957e-1(1.68e-2)-	2.1490e-1(2.51e-2)-	2.3206e-1(3.17e-2)-	1.3651e-1(4.28e-2)-	1.2359e-1(2.31e-2)-	1.0091e-1(2.38e-2)
	ZCAT4	2	8.1048e-3(1.47e-2)=	4.1748e-2(6.02e-2)-	8.8685e-2(7.22e-2)-	4.4013e-2(4.62e-2)-	1.7959e-2(2.26e-2)=	4.2199e-2(4.63e-2)-	3.6389e-2(4.66e-2)
		3	1.7963e-1(3.94e-2)-	1.7947e-1(5.37e-2)-	2.4463e-1(6.47e-2)-	2.1285e-1(9.59e-2)-	1.3978e-1(4.52e-2)=	1.5081e-1(5.10e-2)-	1.3160e-1(4.27e-2)
	ZCAT8	2	1.5840e-2(1.58e-2)-	2.2899e-2(1.73e-2)-	7.8536e-2(3.27e-2)-	2.1272e-2(2.37e-2)-	1.7906e-2(1.91e-2)-	2.3872e-2(1.72e-2)-	1.3986e-2(1.94e-2)
		3	1.4679e-1(3.10e-2)-	8.2637e-2(2.20e-2)+	2.9342e-1(1.29e-1)-	1.9857e-1(6.00e-2)-	1.8493e-1(1.74e-1)-	1.5814e-1(4.33e-2)-	1.0479e-1(4.17e-2)
	ZCAT9	2	1.4689e-2(1.44e-2)=	2.4977e-2(1.29e-2)-	6.6453e-2(3.27e-2)-	2.1776e-2(1.64e-2)-	2.8312e-2(1.95e-2)-	2.4603e-2(2.25e-2)-	1.7409e-2(1.55e-2)
		3	1.3361e-1(2.60e-2)-	1.0168e-1(2.16e-2)-	1.7977e-1(3.55e-2)-	2.3164e-1(1.77e-1)-	9.9748e-2(3.54e-2)=	1.1052e-1(3.55e-2)-	8.9216e-2(2.41e-2)
Flat	ZCAT5	2	2.2709e-2(2.63e-2)+	3.3677e-2(4.87e-2)=	8.5106e-2(4.01e-2)-	3.8524e-2(3.55e-2)=	1.0588e-1(1.15e-1)-	1.8534e-2(2.27e-2)=	3.4794e-2(2.83e-2)
		3	1.0645e-1(1.68e-2)-	8.8322e-2(1.20e-2)-	1.3764e-1(2.46e-2)-	1.3399e-1(1.31e-2)-	1.1636e-1(3.46e-2)-	8.2259e-2(1.05e-2)-	6.7062e-2(1.14e-2)
	ZCAT6	2	2.5452e-2(3.59e-2)=	8.4143e-2(5.05e-2)-	6.0626e-2(4.99e-2)-	5.2109e-2(4.55e-2)-	7.8816e-2(4.44e-2)-	5.1819e-2(5.49e-2)-	2.4683e-2(3.48e-2)
		3	1.2227e-1(1.97e-2)-	8.7679e-2(9.71e-3)-	1.3533e-1(2.56e-2)-	1.2723e-1(1.45e-2)-	1.3143e-1(4.53e-2)-	8.0814e-2(2.47e-2)=	7.6013e-2(1.22e-2)
	ZCAT7	2	1.7103e-2(1.92e-2)+	1.6283e-1(1.43e-1)=	1.2575e-1(1.24e-1)=	7.7283e-2(9.45e-2)=	6.5669e-2(1.30e-1)=	5.2690e-2(9.79e-2)=	1.2668e-1(1.42e-1)
		3	1.4910e-1(4.17e-2)-	9.7630e-2(6.30e-2)=	1.9110e-1(3.31e-2)-	1.6866e-1(3.71e-2)-	1.1592e-1(5.78e-2)=	1.0493e-1(2.25e-2)=	1.2447e-1(7.30e-2)
	ZCAT10	2	1.7121e-2(2.04e-2)=	6.9443e-2(5.34e-2)-	6.9404e-2(4.14e-2)-	2.9498e-2(4.14e-2)=	1.8352e-1(2.44e-1)-	4.0452e-2(3.46e-2)-	2.7869e-2(2.99e-2)
		3	1.1120e-1(1.89e-2)-	8.6011e-2(6.61e-3)=	1.4323e-1(2.16e-2)-	1.5108e-1(2.60e-2)-	1.5873e-1(9.00e-2)-	8.9239e-2(2.13e-2)=	8.2792e-2(1.43e-2)
Disconnected	ZCAT11	2	9.6941e-3(1.14e-2)=	2.7348e-2(4.73e-2)+	8.6258e-2(5.19e-2)-	3.3916e-2(5.05e-2)+	8.1502e-2(1.01e-1)-	2.4654e-2(3.82e-2)+	3.4028e-2(6.42e-2)
		3	1.5419e-1(3.27e-2)-	9.9842e-2(1.23e-2)-	2.0472e-1(3.84e-2)-	2.3046e-1(6.19e-2)-	1.4921e-1(7.25e-2)-	1.1423e-1(3.36e-2)-	8.9837e-2(1.87e-2)
	ZCAT12	2	1.0145e-2(1.14e-2)=	8.3749e-2(1.65e-1)-	9.4720e-2(5.66e-2)-	2.6904e-2(5.66e-2)=	5.5212e-2(8.61e-2)=	2.2241e-2(1.66e-2)=	4.2620e-2(6.96e-2)
		3	1.7108e-1(3.73e-2)-	1.0070e-1(1.23e-2)=	2.7768e-1(3.77e-2)-	2.3603e-1(3.74e-2)-	1.4645e-1(4.33e-2)-	1.2566e-1(2.12e-2)-	9.6159e-2(1.73e-2)
	ZCAT13	2	9.2208e-3(1.35e-2)=	5.7287e-2(1.51e-1)=	9.6784e-2(5.67e-2)-	8.0258e-2(9.62e-2)-	6.3018e-2(7.99e-2)=	2.5595e-2(4.62e-2)=	5.7088e-2(8.54e-2)
		3	1.5473e-1(3.60e-2)-	9.3307e-2(2.42e-2)=	2.1245e-1(4.62e-2)-	2.1211e-1(3.68e-2)-	1.4307e-1(4.76e-2)-	1.0900e-1(3.21e-2)-	9.2133e-2(3.05e-2)
Degenerated	ZCAT14	2	4.2691e-3(2.44e-3)+	1.7846e-2(1.21e-2)-	8.4586e-2(3.37e-2)-	3.2759e-2(2.32e-2)-	1.8769e-2(1.42e-2)=	1.6509e-2(1.32e-2)-	1.1780e-2(1.37e-2)
		3	1.1874e-1(1.96e-2)=	9.3293e-2(1.73e-2)=	2.2542e-1(4.57e-2)-	2.1101e-1(5.74e-2)-	1.8506e-1(4.60e-2)-	1.5816e-1(4.14e-2)-	1.0818e-1(3.50e-2)
Mixed	ZCAT15	2	1.0622e-2(8.03e-3)=	2.6639e-2(6.16e-2)=	9.2197e-2(7.56e-2)-	3.2390e-2(6.95e-2)=	1.0720e-1(1.63e-1)=	3.5554e-2(8.25e-2)=	2.4324e-2(5.66e-2)
		3	2.7036e-1(6.67e-2)-	7.3775e-2(1.07e-1)+	3.0383e-1(1.14e-1)-	4.0443e-1(1.30e-1)-	2.8868e-1(1.10e-1)-	2.0871e-1(6.71e-2)=	1.8745e-1(9.92e-2)
	ZCAT16	2	9.8851e-3(1.19e-2)=	3.7482e-2(6.58e-2)=	7.4078e-2(7.16e-2)-	3.4317e-2(5.76e-2)=	2.7219e-2(4.85e-2)=	1.7421e-2(1.86e-2)=	3.0642e-2(5.54e-2)
		3	1.8994e-1(3.11e-2)-	9.1340e-2(2.07e-2)=	2.1743e-1(1.97e-2)-	2.0592e-1(3.07e-2)-	1.6866e-1(2.59e-2)-	1.3946e-1(5.62e-2)-	9.8513e-2(2.60e-2)
	ZCAT17	2	1.4319e-2(2.24e-2)+	2.0738e-2(2.19e-2)=	7.8639e-2(5.76e-2)-	4.6477e-2(3.37e-2)-	1.6534e-1(1.27e-1)-	1.3721e-2(1.65e-2)=	2.7871e-2(2.87e-2)
		3	9.6999e-2(1.11e-2)-	6.6867e-2(1.05e-2)-	1.4756e-1(1.79e-2)-	1.3865e-1(1.96e-2)-	1.4180e-1(8.02e-2)-	6.9345e-2(1.16e-2)-	5.3801e-2(9.41e-3)
	ZCAT18	2	1.5675e-2(1.97e-2)+	1.9302e-1(1.39e-1)=	1.3348e-1(1.19e-1)=	9.6365e-2(1.22e-1)=	1.2251e-1(1.64e-1)=	5.9206e-2(8.79e-2)=	1.5723e-1(1.50e-1)
		3	1.4078e-1(4.03e-2)-	7.2605e-2(3.15e-2)+	1.6810e-1(2.93e-2)-	1.4669e-1(4.34e-2)-	1.2061e-1(6.68e-2)=	8.6706e-2(2.37e-2)=	9.4246e-2(4.08e-2)
	ZCAT19	2	1.2000e-2(1.33e-2)+	2.1551e-2(1.70e-2)=	8.6638e-2(4.36e-2)-	1.9104e-2(1.30e-2)=	2.3164e-2(1.55e-2)=	1.8822e-2(1.61e-2)=	2.0734e-2(1.77e-2)
		3	1.8160e-1(3.56e-2)-	1.1628e-1(1.41e-2)=	2.3998e-1(3.64e-2)-	2.1003e-1(3.47e-2)-	1.3570e-1(2.97e-2)-	1.2118e-1(3.00e-2)=	1.1411e-1(2.35e-2)
	ZCAT20	2	1.5458e-2(1.96e-2)+	1.4517e-1(1.40e-1)=	1.9014e-1(1.62e-1)-	1.5104e-1(1.48e-1)=	2.1186e-1(2.06e-1)-	4.3466e-2(6.96e-2)=	1.3678e-1(1.58e-1)
		3	1.1732e-1(2.99e-2)=	9.2870e-2(9.03e-2)=	1.9944e-1(6.43e-2)=	1.5522e-1(6.87e-2)=	1.5537e-1(1.24e-1)=	8.8630e-2(3.84e-2)=	1.9121e-1(1.60e-1)
+/ − / ≈			8/18/14	4/18/18	0/37/3	2/27/11	0/25/15	2/20/18	

Table A.13
HV values obtained by TS-MOEA-EPF and 7 of 13 MaOEAs on ZCAT, where "+", "-", and "=" respectively represent the comparison algorithms being superior, inferior, and equivalent to TS-MOEA-EPF (Part I)

Property	Problem	M	CA-MOEA	DEA-GNG	RVEA-iGNG	PeEA	LMPFE	HEA	RVEA-AdaAC	TS-MOEA-EPF
Regular	ZCAT1	2	5.7631e-1(9.66e-3)-	5.6986e-1(1.71e-2)-	5.6849e-1(1.38e-2)-	5.6686e-1(1.01e-2)-	5.6421e-1(1.43e-2)-	5.5830e-1(2.75e-2)-	5.2869e-1(1.86e-2)-	5.7669e-1(1.40e-2)
		3	7.1414e-1(2.28e-2)-	7.2433e-1(1.72e-2)-	7.4374e-1(2.31e-2)=	7.1060e-1(1.19e-2)-	7.2297e-1(2.55e-2)-	7.0255e-1(6.72e-2)-	7.0914e-1(2.63e-2)-	7.4133e-1(2.17e-2)
	ZCAT2	2	8.1228e-1(7.85e-3)=	8.1056e-1(1.13e-2)=	7.9417e-1(2.22e-2)-	8.0099e-1(1.06e-2)=	8.0839e-1(1.05e-2)=	7.9395e-1(1.87e-2)-	7.7799e-1(1.87e-2)-	8.0433e-1(1.52e-2)
		3	9.4758e-1(1.91e-2)=	9.3660e-1(1.72e-2)-	9.6115e-1(1.33e-2)+	9.1688e-1(2.11e-2)-	9.4871e-1(1.96e-2)=	9.3426e-1(2.79e-2)-	9.3513e-1(2.12e-2)-	9.5307e-1(1.59e-2)
Partial	ZCAT3	2	5.7561e-1(9.87e-3)+	5.7147e-1(1.48e-2)-	5.6179e-1(2.49e-2)-	5.6736e-1(9.86e-3)-	5.6965e-1(1.16e-2)-	5.6199e-1(1.98e-2)-	5.2398e-1(3.02e-2)-	5.7531e-1(1.21e-2)
		3	3.8528e-1(2.71e-2)-	4.0528e-1(5.73e-2)-	4.7991e-1(1.22e-2)=	4.5949e-1(1.13e-2)-	4.1773e-1(1.93e-2)-	4.6507e-1(1.33e-2)-	3.3562e-1(2.79e-2)-	4.7351e-1(1.62e-2)
	ZCAT4	2	5.7384e-1(2.99e-2)=	5.6235e-1(3.57e-2)+	4.2437e-1(9.18e-2)-	5.4786e-1(3.58e-2)-	5.3289e-1(4.74e-2)-	3.8884e-1(9.71e-2)-	5.4158e-1(4.64e-2)-	5.5413e-1(3.87e-2)
		3	4.4872e-1(1.92e-2)-	4.8896e-1(4.19e-2)-	4.5811e-1(5.26e-2)-	4.8444e-1(3.31e-2)-	4.5612e-1(3.63e-2)-	4.6962e-1(4.28e-2)-	4.0351e-1(4.26e-2)-	5.0482e-1(2.89e-2)
	ZCAT8	2	3.4310e-1(6.56e-3)+	3.3582e-1(1.56e-2)-	3.2112e-1(2.83e-2)-	3.2682e-1(1.64e-2)-	3.2044e-1(2.68e-2)-	3.0982e-1(6.14e-2)-	2.7850e-1(2.58e-2)-	3.3979e-1(1.40e-2)
		3	2.5549e-2(6.72e-3)-	2.3951e-2(1.34e-2)-	2.5065e-2(9.48e-3)-	2.0528e-2(1.12e-2)-	3.1906e-2(9.81e-3)-	5.7840e-2(4.99e-3)=	1.0498e-2(5.12e-3)-	5.9936e-2(7.16e-3)
	ZCAT9	2	3.3908e-1(9.72e-3)=	3.3157e-1(1.52e-2)-	3.1446e-1(3.78e-2)-	3.2527e-1(1.40e-2)-	3.3042e-1(1.47e-2)-	3.2738e-1(2.06e-2)-	2.8284e-1(2.20e-2)-	3.3736e-1(1.10e-2)
		3	2.0313e-1(2.03e-2)-	2.0890e-1(5.58e-2)-	2.5371e-1(3.45e-2)=	2.3868e-1(1.12e-2)-	2.1732e-1(1.95e-2)-	2.5746e-1(1.13e-2)=	1.4461e-1(2.33e-2)-	2.5952e-1(1.43e-2)
Flat	ZCAT5	2	8.8830e-1(7.95e-3)=	8.8296e-1(1.09e-2)=	8.7442e-1(1.92e-2)=	8.7568e-1(1.14e-2)=	8.8844e-1(7.63e-3)=	8.4829e-1(3.27e-2)-	8.6216e-1(1.47e-2)-	8.7954e-1(1.29e-2)
		3	8.5548e-1(1.82e-2)-	8.8084e-1(1.47e-2)-	9.1442e-1(2.27e-2)=	9.0022e-1(1.37e-2)-	8.5419e-1(2.76e-2)-	8.7917e-1(3.48e-2)-	8.7108e-1(3.65e-2)-	9.1280e-1(1.60e-2)
	ZCAT6	2	5.7991e-1(7.56e-3)=	5.7512e-1(1.25e-2)-	5.5625e-1(2.01e-2)-	5.7159e-1(1.22e-2)-	5.7042e-1(1.43e-2)-	5.4655e-1(2.36e-2)-	5.3336e-1(2.15e-2)-	5.7761e-1(1.20e-2)
		3	4.5200e-1(2.26e-2)-	5.1606e-1(2.00e-2)-	5.3603e-1(1.89e-2)-	5.3388e-1(1.50e-2)-	4.8115e-1(2.72e-2)-	5.4506e-1(2.07e-2)=	4.4804e-1(2.74e-2)-	5.4572e-1(1.70e-2)
	ZCAT7	2	5.7498e-1(1.78e-2)=	5.5707e-1(2.73e-2)=	5.2315e-1(4.12e-2)-	5.6901e-1(1.07e-2)=	5.6261e-1(2.43e-2)=	4.8919e-1(4.34e-2)-	5.3289e-1(2.27e-2)-	5.4867e-1(3.71e-2)
		3	4.8192e-1(1.76e-2)-	5.0082e-1(2.55e-2)-	5.1943e-1(1.62e-2)-	4.9012e-1(1.61e-2)-	4.9025e-1(2.15e-2)-	5.1822e-1(1.91e-2)-	4.4678e-1(2.06e-2)-	5.2817e-1(2.07e-2)
	ZCAT10	2	2.1296e-1(9.94e-3)+	2.0984e-1(1.14e-2)-	1.9164e-1(3.59e-2)-	1.9555e-1(1.48e-2)-	2.0448e-1(1.60e-2)-	1.2842e-1(7.51e-2)-	1.5948e-1(2.20e-2)-	2.1214e-1(1.12e-2)
		3	8.8897e-2(5.82e-3)-	1.0315e-1(7.73e-3)-	9.0398e-2(8.20e-3)-	8.6603e-2(1.14e-2)-	9.8648e-2(5.87e-3)-	1.0460e-1(5.70e-3)-	8.5707e-2(7.02e-3)-	1.1257e-1(5.25e-3)
Disconnected	ZCAT11	2	5.4262e-1(9.52e-3)=	5.3843e-1(1.98e-2)=	5.0614e-1(3.32e-2)-	5.3577e-1(8.43e-3)-	5.2575e-1(2.12e-2)-	4.9436e-1(4.38e-2)-	4.9676e-1(1.81e-2)-	5.3827e-1(1.79e-2)
		3	3.8117e-1(1.95e-2)-	4.0076e-1(4.76e-2)-	4.6283e-1(1.55e-2)=	4.3799e-1(1.03e-2)-	4.1105e-1(2.30e-2)-	4.5649e-1(1.67e-2)=	3.2802e-1(2.04e-2)-	4.5854e-1(1.24e-2)
	ZCAT12	2	5.8133e-1(1.06e-2)=	5.8020e-1(1.29e-2)=	5.4932e-1(4.73e-2)-	5.7290e-1(1.07e-2)=	5.7286e-1(1.78e-2)=	5.1749e-1(7.51e-2)-	5.3284e-1(2.17e-2)-	5.7290e-1(1.89e-2)
		3	1.4519e-1(1.66e-2)-	1.6990e-1(2.52e-2)-	2.1411e-1(1.21e-2)-	1.8424e-1(1.52e-2)-	1.6961e-1(3.49e-2)-	1.9963e-1(9.67e-3)-	7.0430e-2(2.41e-2)-	2.2061e-1(8.02e-3)
	ZCAT13	2	6.7271e-1(1.32e-2)=	6.6889e-1(1.88e-2)=	6.4802e-1(3.46e-2)-	6.7005e-1(1.13e-2)=	6.6004e-1(2.84e-2)=	6.3026e-1(3.30e-2)-	6.3718e-1(1.97e-2)-	6.6109e-1(3.16e-2)
		3	5.5395e-1(1.73e-2)-	5.7836e-1(4.25e-2)-	6.3027e-1(2.42e-2)=	6.0675e-1(1.22e-2)-	5.7330e-1(2.64e-2)-	6.3310e-1(1.59e-2)=	5.0757e-1(1.02e-1)-	6.3445e-1(1.62e-2)
Degenerated	ZCAT14	2	5.7560e-1(1.60e-2)+	5.6207e-1(3.17e-2)+	5.1757e-1(4.52e-2)+	5.7022e-1(8.60e-3)+	5.6921e-1(1.49e-2)+	4.2013e-1(2.42e-2)-	5.2667e-1(3.03e-2)+	4.3864e-1(1.10e-2)
		3	4.8477e-1(1.45e-2)+	5.0311e-1(2.05e-2)+	5.1064e-1(2.28e-2)+	4.5786e-1(2.05e-2)+	4.9025e-1(2.34e-2)+	3.5517e-1(8.70e-3)+	4.3952e-1(2.97e-2)+	3.4452e-1(1.02e-2)
Mixed	ZCAT15	2	6.0396e-1(7.93e-3)=	6.0254e-1(1.29e-2)=	5.8596e-1(2.22e-2)-	5.8915e-1(8.01e-3)-	5.9521e-1(1.58e-2)=	5.5997e-1(3.70e-2)-	5.6750e-1(1.13e-2)-	5.9841e-1(1.25e-2)
		3	3.4103e-1(3.42e-2)-	4.0292e-1(1.88e-2)-	4.1464e-1(2.34e-2)-	3.8299e-1(5.11e-2)-	4.0041e-1(1.65e-2)-	4.5136e-1(5.97e-3)+	3.3373e-1(3.12e-2)-	4.4307e-1(1.08e-2)
	ZCAT16	2	4.8727e-1(1.38e-2)=	4.7887e-1(1.38e-2)-	4.6586e-1(2.52e-2)-	4.7575e-1(1.19e-2)-	4.7299e-1(1.56e-2)-	4.5353e-1(4.31e-2)-	4.3081e-1(2.53e-2)-	4.8181e-1(1.32e-2)
		3	2.7018e-1(2.58e-2)-	3.4156e-1(1.95e-2)-	3.8617e-1(1.20e-2)-	3.7440e-1(1.19e-2)-	3.2186e-1(1.42e-2)-	4.0367e-1(6.64e-3)+	2.8433e-1(2.53e-2)-	3.9348e-1(9.25e-3)
	ZCAT17	2	8.8931e-1(1.01e-2)=	8.7906e-1(1.40e-2)=	8.7163e-1(1.84e-2)-	8.7358e-1(1.09e-2)-	8.8471e-1(1.07e-2)=	8.6220e-1(1.94e-2)-	8.5095e-1(1.53e-2)-	8.8295e-1(1.27e-2)
		3	7.7531e-1(3.87e-2)-	8.5773e-1(1.80e-2)-	8.7087e-1(3.87e-2)=	8.6398e-1(2.40e-2)-	8.1260e-1(2.42e-2)-	8.8064e-1(2.53e-2)=	8.2003e-1(4.12e-2)-	8.8209e-1(1.80e-2)
	ZCAT18	2	5.7063e-1(1.93e-2)+	5.5101e-1(3.31e-2)=	5.1720e-1(3.97e-2)-	5.6866e-1(1.00e-2)+	5.6033e-1(2.49e-2)=	4.8803e-1(2.30e-2)-	5.2298e-1(2.51e-2)=	5.3729e-1(3.89e-2)
		3	4.5521e-1(2.24e-2)-	4.8651e-1(1.97e-2)-	5.0659e-1(1.81e-2)=	4.5175e-1(5.41e-2)-	4.7172e-1(1.80e-2)-	5.1510e-1(1.53e-2)=	4.2594e-1(2.85e-2)-	5.1323e-1(1.57e-2)
	ZCAT19	2	5.7387e-1(1.12e-2)=	5.7688e-1(1.04e-2)=	5.5785e-1(1.80e-2)-	5.6181e-1(1.21e-2)-	5.6667e-1(1.38e-2)=	5.6133e-1(1.89e-2)=	5.3640e-1(2.49e-2)-	5.7005e-1(1.35e-2)
		3	4.0272e-1(2.05e-2)-	4.4691e-1(2.00e-2)-	4.7937e-1(1.62e-2)=	4.5785e-1(1.21e-2)-	4.1557e-1(1.93e-2)-	4.7749e-1(1.17e-2)=	3.6955e-1(2.43e-2)-	4.7849e-1(1.41e-2)
	ZCAT20	2	5.7340e-1(1.57e-2)=	5.6934e-1(2.23e-2)=	5.0340e-1(3.73e-2)-	5.6242e-1(1.52e-2)=	5.6555e-1(2.55e-2)=	5.0617e-1(4.21e-2)-	5.2537e-1(1.90e-2)-	5.4497e-1(4.10e-2)
		3	4.9134e-1(1.58e-2)-	5.0193e-1(2.65e-2)-	5.1427e-1(2.12e-2)=	4.7343e-1(9.06e-2)-	4.7603e-1(2.76e-2)-	5.1539e-1(2.17e-2)=	4.2354e-1(2.89e-2)-	5.2441e-1(1.66e-2)
+/ − / ≈			6/19/15	3/26/11	3/26/11	3/31/6	2/27/11	3/27/10	2/37/1	

Table A.14
HV values obtained by TS-MOEA-EPF and 6 of 13 MaOEAs on ZCAT, where "+", "-", and "=" respectively represent the comparison algorithms being superior, inferior, and equivalent to TS-MOEA-EPF (Part II)

Property	Problem	M	AdaW	MultiGPO	TS-NSGA-II	AR-MOEA	hpaEA	CARV-MOEA	TS-MOEA-EPF
Regular	ZCAT1	2	5.7498e-1(1.19e-2)-	5.6902e-1(1.15e-2)-	5.3450e-1(3.21e-2)-	5.6814e-1(1.56e-2)-	5.6854e-1(1.23e-2)-	5.7009e-1(1.28e-2)-	5.7669e-1(1.40e-2)
		3	7.3893e-1(1.65e-2)=	7.3433e-1(1.09e-2)-	7.2542e-1(2.04e-2)-	7.2795e-1(2.50e-2)-	7.3729e-1(1.89e-2)=	6.9308e-1(2.40e-2)-	7.4133e-1(2.17e-2)
	ZCAT2	2	8.1560e-1(7.92e-3)=	8.0190e-1(1.34e-2)=	7.3046e-1(8.07e-2)-	8.0137e-1(1.47e-2)=	7.8407e-1(2.67e-2)-	8.0125e-1(1.28e-2)-	8.0433e-1(1.52e-2)
		3	9.5534e-1(1.76e-2)=	9.3477e-1(1.61e-2)-	9.2785e-1(2.85e-2)-	9.4552e-1(1.52e-2)=	9.4221e-1(1.55e-2)-	8.8620e-1(1.77e-2)-	9.5307e-1(1.59e-2)
Partial	ZCAT3	2	5.7591e-1(9.39e-3)=	5.6565e-1(1.29e-2)-	5.0070e-1(4.54e-2)-	5.6683e-1(1.43e-2)-	5.6391e-1(2.00e-2)-	5.6623e-1(1.26e-2)-	5.7531e-1(1.21e-2)
		3	4.3713e-1(2.06e-2)-	4.6345e-1(1.52e-2)-	3.9505e-1(1.65e-2)-	3.8443e-1(1.86e-2)-	4.5753e-1(2.18e-2)-	4.4380e-1(1.63e-2)-	4.7351e-1(1.62e-2)
	ZCAT4	2	5.7864e-1(1.35e-2)=	5.6139e-1(3.02e-2)=	4.9613e-1(7.38e-2)-	5.4690e-1(3.96e-2)-	5.7225e-1(1.68e-2)=	5.4820e-1(4.00e-2)-	5.5413e-1(3.87e-2)
		3	4.7429e-1(2.70e-2)-	4.6117e-1(4.14e-2)-	4.4387e-1(3.97e-2)-	4.5713e-1(4.73e-2)-	5.0495e-1(2.40e-2)=	4.7462e-1(3.47e-2)-	5.0482e-1(2.89e-2)
	ZCAT8	2	3.3799e-1(1.12e-2)-	3.3080e-1(1.37e-2)-	2.7956e-1(3.17e-2)-	3.1674e-1(7.64e-2)-	3.3659e-1(1.35e-2)-	3.3214e-1(1.27e-2)-	3.3979e-1(1.40e-2)
		3	4.3285e-2(6.09e-3)-	5.8675e-2(3.80e-3)=	2.5460e-2(6.28e-3)-	3.5827e-2(6.13e-3)-	4.8206e-2(1.31e-2)-	4.1668e-2(7.95e-3)-	5.9936e-2(7.16e-3)
	ZCAT9	2	3.3881e-1(1.06e-2)=	3.3035e-1(1.03e-2)-	2.9037e-1(2.97e-2)-	3.3370e-1(1.17e-2)-	3.2903e-1(1.39e-2)-	3.3172e-1(1.64e-2)-	3.3736e-1(1.10e-2)
		3	2.2845e-1(1.65e-2)-	2.5494e-1(1.55e-2)=	2.0324e-1(2.11e-2)-	1.8932e-1(2.84e-2)-	2.5543e-1(1.90e-2)=	2.4752e-1(2.10e-2)-	2.5952e-1(1.43e-2)
Flat	ZCAT5	2	8.8490e-1(1.20e-2)=	8.7804e-1(2.38e-2)=	8.3586e-1(2.51e-2)-	8.7838e-1(1.60e-2)=	8.5729e-1(3.65e-2)-	8.8471e-1(1.08e-2)=	8.7954e-1(1.29e-2)
		3	8.9187e-1(2.16e-2)-	9.1228e-1(1.87e-2)=	8.8166e-1(2.88e-2)-	8.9018e-1(1.97e-2)-	8.9056e-1(2.91e-2)-	8.6011e-1(1.56e-2)-	9.1280e-1(1.60e-2)
	ZCAT6	2	5.7695e-1(1.24e-2)=	5.5154e-1(2.13e-2)-	5.6282e-1(1.84e-2)-	5.6790e-1(1.57e-2)-	5.5955e-1(1.43e-2)-	5.6775e-1(2.00e-2)-	5.7761e-1(1.20e-2)
		3	5.0646e-1(1.88e-2)-	5.1546e-1(1.42e-2)-	4.8514e-1(2.74e-2)-	5.0723e-1(2.24e-2)-	5.2322e-1(2.40e-2)-	5.2506e-1(2.69e-2)-	5.4572e-1(1.70e-2)
	ZCAT7	2	5.7566e-1(1.02e-2)=	5.3556e-1(3.72e-2)-	5.2783e-1(2.99e-2)=	5.5404e-1(2.72e-2)=	5.6371e-1(3.32e-2)=	5.6386e-1(2.62e-2)=	5.4867e-1(3.71e-2)
		3	5.0219e-1(2.06e-2)-	5.3204e-1(1.45e-2)=	4.7313e-1(1.91e-2)-	4.8255e-1(2.54e-2)-	5.1247e-1(2.29e-2)-	4.9165e-1(1.79e-2)-	5.2817e-1(2.07e-2)
	ZCAT10	2	2.1532e-1(7.59e-3)=	1.9342e-1(2.23e-2)-	1.9431e-1(1.70e-2)-	2.1116e-1(1.53e-2)=	1.7015e-1(6.40e-2)-	2.0698e-1(1.29e-2)-	2.1214e-1(1.12e-2)
		3	1.0449e-1(7.57e-3)-	1.1528e-1(3.70e-3)+	1.0135e-1(6.17e-3)-	9.9500e-2(6.57e-3)-	1.0494e-1(9.76e-3)-	1.1345e-1(5.48e-3)=	1.1257e-1(5.25e-3)
Disconnected	ZCAT11	2	5.4455e-1(1.01e-2)=	5.3891e-1(1.38e-2)=	4.8605e-1(2.78e-2)-	5.3338e-1(1.43e-2)-	5.2179e-1(2.70e-2)-	5.3624e-1(1.36e-2)-	5.3827e-1(1.79e-2)
		3	4.1702e-1(1.96e-2)-	4.4617e-1(1.10e-2)-	3.8851e-1(2.35e-2)-	3.7213e-1(3.33e-2)-	4.3975e-1(2.04e-2)-	4.2986e-1(2.04e-2)-	4.5854e-1(1.24e-2)
	ZCAT12	2	5.8181e-1(1.11e-2)=	5.5368e-1(4.94e-2)-	5.0999e-1(3.61e-2)-	5.7402e-1(1.98e-2)=	5.6810e-1(2.66e-2)=	5.6977e-1(1.62e-2)=	5.7290e-1(1.89e-2)
		3	1.7754e-1(1.51e-2)-	2.1090e-1(7.67e-3)-	1.2660e-1(1.23e-2)-	1.4018e-1(1.53e-2)-	2.0041e-1(1.55e-2)-	1.9747e-1(9.94e-3)-	2.2061e-1(8.02e-3)
	ZCAT13	2	6.7918e-1(1.17e-2)=	6.6393e-1(2.07e-2)=	5.9484e-1(5.11e-2)-	6.5231e-1(3.42e-2)-	6.5680e-1(2.76e-2)=	6.6928e-1(1.89e-2)=	6.6109e-1(3.16e-2)
		3	5.9156e-1(2.15e-2)-	6.2210e-1(1.72e-2)-	5.6362e-1(2.39e-2)-	5.5960e-1(2.05e-2)-	6.1457e-1(1.84e-2)-	6.1024e-1(1.60e-2)-	6.3445e-1(1.62e-2)
Degenerated	ZCAT14	2	5.7531e-1(1.10e-2)+	4.3034e-1(1.24e-2)-	3.6756e-1(3.30e-2)-	4.2151e-1(1.80e-2)-	4.3155e-1(1.26e-2)-	4.3489e-1(1.12e-2)-	4.3864e-1(1.10e-2)
		3	5.0571e-1(1.57e-2)+	3.5056e-1(5.79e-3)+	3.0496e-1(1.38e-2)-	3.0876e-1(1.62e-2)-	3.2623e-1(1.28e-2)-	3.2250e-1(1.14e-2)-	3.4452e-1(1.02e-2)
Mixed	ZCAT15	2	6.0283e-1(7.43e-3)=	5.9742e-1(8.45e-3)=	5.3615e-1(5.20e-2)-	5.9540e-1(1.67e-2)=	5.8707e-1(2.79e-2)=	5.9286e-1(1.63e-2)=	5.9841e-1(1.25e-2)
		3	4.0199e-1(1.83e-2)-	4.6293e-1(6.88e-3)+	3.9957e-1(2.13e-2)-	3.8685e-1(2.11e-2)-	4.2451e-1(9.90e-3)-	4.3150e-1(1.14e-2)-	4.4307e-1(1.08e-2)
	ZCAT16	2	4.8729e-1(1.14e-2)=	4.5451e-1(1.08e-1)-	4.4669e-1(2.81e-2)-	4.7838e-1(1.57e-2)-	4.7863e-1(1.33e-2)=	4.8014e-1(1.76e-2)=	4.8181e-1(1.32e-2)
		3	3.4242e-1(1.75e-2)-	3.9230e-1(8.88e-3)=	3.3945e-1(8.82e-3)-	3.4084e-1(1.45e-2)-	3.7155e-1(1.07e-2)-	3.7267e-1(2.13e-2)-	3.9348e-1(9.25e-3)
	ZCAT17	2	8.8876e-1(1.00e-2)=	8.8432e-1(8.97e-3)=	8.4315e-1(3.45e-2)-	8.7514e-1(1.48e-2)-	8.4059e-1(3.96e-2)-	8.8655e-1(7.76e-3)=	8.8295e-1(1.27e-2)
		3	8.5528e-1(1.87e-2)-	8.8169e-1(1.48e-2)=	8.5119e-1(2.87e-2)-	8.3817e-1(2.81e-2)-	8.6358e-1(2.80e-2)-	8.6081e-1(2.13e-2)-	8.8209e-1(1.80e-2)
	ZCAT18	2	5.7668e-1(1.04e-2)+	5.2853e-1(3.68e-2)=	5.2360e-1(3.44e-2)=	5.5255e-1(3.17e-2)=	5.4856e-1(4.20e-2)=	5.5999e-1(2.31e-2)=	5.3729e-1(3.89e-2)
		3	4.9206e-1(2.10e-2)-	5.1969e-1(1.10e-2)=	4.6971e-1(1.93e-2)-	4.6179e-1(2.72e-2)-	5.0202e-1(1.90e-2)-	4.9755e-1(1.79e-2)-	5.1323e-1(1.57e-2)
	ZCAT19	2	5.7638e-1(1.01e-2)=	5.6362e-1(1.42e-2)-	4.9864e-1(4.38e-2)-	5.7123e-1(9.92e-3)=	5.6941e-1(1.02e-2)=	5.7069e-1(1.21e-2)=	5.7005e-1(1.35e-2)
		3	4.3995e-1(2.10e-2)-	4.5888e-1(1.22e-2)-	4.1061e-1(2.01e-2)-	4.1764e-1(1.87e-2)-	4.7363e-1(1.36e-2)=	4.6445e-1(1.53e-2)-	4.7849e-1(1.41e-2)
	ZCAT20	2	5.7687e-1(1.02e-2)+	5.4327e-1(3.56e-2)=	5.1543e-1(4.17e-2)-	5.3871e-1(3.63e-2)=	5.1546e-1(5.29e-2)-	5.7004e-1(1.26e-2)=	5.4497e-1(4.10e-2)
		3	5.0480e-1(1.74e-2)-	5.2833e-1(1.46e-2)=	4.7031e-1(2.29e-2)-	4.8461e-1(1.81e-2)-	5.0943e-1(2.09e-2)-	4.9962e-1(2.13e-2)-	5.2441e-1(1.66e-2)
+/ − / ≈			4/19/17	3/20/17	0/38/2	0/30/10	0/28/12	0/29/11	

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